

Review

Navigating Challenges and Harnessing Opportunities: Deep Learning Applications in Internet of Medical Things

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Abstract: Integrating deep learning (DL) with the Internet of Medical Things (IoMT) is a paradigm shift in modern healthcare, offering enormous opportunities for patient care, diagnostics, and treatment. Implementing DL with IoMT has the potential to deliver better diagnosis, treatment, and patient management. However, the practical implementation has challenges, including data quality, privacy, interoperability, and limited computational resources. This survey article provides a conceptual IoMT framework for healthcare, synthesizes and identifies the state-of-the-art solutions that tackle the challenges of the current applications of DL, and analyzes existing limitations and potential future developments. Through an analysis of case studies and real-world implementations, this work provides insights into best practices and lessons learned, including the importance of robust data preprocessing, integration with legacy systems, and human-centric design. Finally, we outline future research directions, emphasizing the development of transparent, scalable, and privacy-preserving DL models to realize the full potential of IoMT in healthcare. This survey aims to serve as a foundational reference for researchers and practitioners seeking to navigate the challenges and harness the opportunities in this rapidly evolving field.



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1. Introduction

Due to the high cost of upgrades, complex regulations, and complicated maintenance procedures, outdated systems and practices still exist in healthcare, such as the electrical health record (EHR) system. The EHR system was introduced in the 1970s, and in the year 2009, the federal government set aside USD 27 billion to encourage hospitals and providers to adopt the EHR system [1]. According to The National Electronic Health Records Survey (NEHRS) from the Centers for Disease Control and Prevention (CDC), by the year 2021, 88% of U.S. office-based physicians had adopted the EHR system [2]. That means after 12 years, many healthcare facilities still rely on paper records rather than EHRs.

The slow adoption of the EHR system can be attributed to several factors, including the high implementation cost, resistance to change, incomplete records, and the lack of a common standard across different datasets. Among these challenges, the most significant barrier is data collection [3]. For instance, even though the vision of cyber-physical systems (CPSs) adopts the layered vertical design architecture to integrate cyberspace and

physical space and synthesizes sensing, communication, computing, and machine learning, supporting real-time monitoring, control, and management of a variety of physical things in various application domains (energy, transportation, manufacturing, homes, healthcare, etc.) [4–17], in smart healthcare CPSs, missing real-time monitoring can lead to delayed responses to critical health events (e.g., heart attacks, strokes), which could otherwise be prevented with early intervention. Manual infusion pump operations increase the risk of human errors, such as incorrect medication dosing, potentially leading to adverse drug events. Similarly, a lack of precision and real-time feedback in surgical procedures may result in complications, prolonged recovery times, or suboptimal outcomes.

In recent decades, technological advancements have transformed healthcare by enhancing diagnostics, treatments, and disease monitoring [18]. Innovations like EHRs, telemedicine [19], deep learning (DL) [20], the Internet of Things (IoT) [21], and robotics [22] have made care more efficient and accessible. Particularly, DL enhances medical cyber-physical systems (MCPSs) with AI, enabling patient empowerment through wearables and mobile health apps for self-monitoring. This transition fosters proactive, patient-centered healthcare driven by advanced sensing and data analytics [23]. DL excels at detecting diseases from MRI, CT, and X-ray images in medical imaging. It addresses challenges like large gigapixel histopathology scans and noisy 3D images, where poor labeling complicates training [24]. DL models analyze ECGs for early arrhythmia signs in smartwatches and fitness trackers, enhancing health monitoring and improving predictions based on heart rate and blood pressure [25].

However, DL models in MCPSs face significant challenges due to limitations in sensing, communication, control, and computing. High-dimensional data from IoMT systems often degrade their real-world performance, leading to data imbalance, noise, and mislabeling issues. Critical resources like processing power and memory are frequently limited, hindering intensive algorithm execution. Additionally, data problems are worsened by fragmentation and incompleteness, including privacy, security, and the availability of well-labeled datasets [26]. DL models also struggle with generalization and bias, performing well in specific contexts while failing in others [27]. The lack of interpretability, where models act as ‘black boxes’, continues to weaken trust among healthcare professionals [28]. Furthermore, integration with healthcare systems poses interoperability challenges for EHRs, potentially disrupting workflows. In high-pressure medical settings, models need to deliver real-time predictions, as the IoMT and DL can automate critical tasks like patient monitoring and drug dispensing. Ultimately, predictive analytics from IoMT data can significantly enhance resource allocation in areas like surgery and ICU.

In this context, the Internet of Medical Things (IoMT) is a transformative ecosystem that integrates medical devices, software, hardware platforms, and networks to collect, process, and share healthcare-related data. This interconnected ecosystem not only enables efficient and real-time healthcare delivery but also supports personalized treatment by leveraging advanced technologies, including sensors, communication networks, edge devices, and cloud platforms. Together, these components establish the backbone of the IoMT’s architecture, driving a patient-centric approach to modern healthcare systems.

Integrating deep learning (DL) with the IoMT [29] will allow enormous data flows, such as high-resolution medical images and real-time data collected by sensors. The data need to be transformed into usable insights to enhance health service quality, scalability, and accessibility. Transmitting raw data across network systems can lead to traffic congestion, data overflow, long latency, etc. DL models can be deployed to locally process data, perform preliminary analyses, detect abnormalities, and transmit only the results instead of the entire dataset. This reduces network load and ensures efficient data transmission without compromising the system’s performance. We reviewed related papers through Google

Scholar, IEEE, and MDPI. We focused on investigating the performance of DL applications in smart healthcare from the IoMT perspective. Additionally, we classified them based on the three dimensions of the problem space and the structured framework. Furthermore, we explored the challenges and opportunities regarding DL in the IoMT.

Table 1 compares our survey and the other six related survey papers. Most of these surveys focus on the application layer, particularly on challenges, especially security. In fact, DL in the network layer and the perception layer is a critical component of IoMT systems. Furthermore, most of these surveys offer only a brief introduction to the DL environment despite its significance as a foundational factor in designing and implementing effective DL models. This gap highlights the necessity of addressing DL in IoMT systems from various angles in our defined problem space.

Table 1. Related surveys on IoMT systems.

IoMT Architecture Layers				Device Deployment			Applications				
Ref.	Perception	Network	App	Home	Wearable	Hospital	Precision	Risk	Personal	Challenges	Year
[30]	Yes	Yes	Yes	No	No	No	Partial	Yes	No	Yes	2022
[31]	Partial	No	Yes	No	No	No	Yes	Yes	No	Yes	2017
[32]	Yes	Yes	Yes	Partial	Partial	Yes	Yes	Yes	No	Yes	2024
[33]	Yes	Yes	Yes	Partial	Yes	Yes	Yes	Yes	Partial	Yes	2021
[34]	Yes	Partial	Yes	No	No	No	Partial	Yes	Partial	Yes	2022
Ours	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	2025

To summarize, the major contributions of this survey paper are as follows:

- Proposed problem space and framework: We present a comprehensive problem space framework that systematically classifies DL-based application scenarios in the IoMT with healthcare applications based on analyzing three orthogonal approaches.
- Comprehensive survey of applications: We review a large number of applications of DL in the IoMT, including precision diagnostics, intelligent surgical tools, wearable devices for health care, and individualized therapy.
- Challenges and future directions: We summarize some current challenges in DL applications for the IoMT, and then show a related future research plan.

The remainder of the article is organized as follows. Section 2 briefly provides some background. Section 3 explains the main benefits and some challenges of using DL for IoMT in healthcare and the architecture of the IoMT system. Section 4 reviews the DL applied in IoMT based on our designed 3D problem space. Section 5 summarizes some existing challenges and issues in IoMT with DL applications, especially in the security of applications. Section 6 outlines some future research work to enhance the framework. Finally, Section 7 concludes the paper.

2. Preliminaries

2.1. Cyber–Physical Systems (CPSs)

A CPS is a system managed and controlled by the computing core with the support of communication networks [35]. A CPS is a system that controls and monitors physical processes by a computing-based system; CPSs have been widely used in a variety of industries, such as smart healthcare, smart cities, smart grids, smart greenhouses, smart industries, and even digital twins [7,36–39]. With advanced sensor technology, high-performance network systems, expanded data storage, and devices integrated into Internet-based systems for remote control, real-time monitoring, and predictive analytics [40], CPSs enable automated manufacturing in smart factories and are used in smart cities [41] for

traffic management and environmental monitoring. For example, interconnected traffic signals [42] adapt in real time to reduce congestion, while environmental sensors monitor temperature and humidity for plant growth.

A typical CPS comprises three main components: physical space, cyberspace, and applications. In the physical space, sensors are attached to the physical objects to retrieve related information. For instance, a temperature and humidity sensor (DTH11) obtains real-time data that are essential to maintain specific environmental conditions for storing physical objects. Actuators are deployed to enable the control of physical objects by receiving messages from the computing system and transmitting the control command to physical devices [35]. In cyberspace, various data are collected and then processed to make decisions. Thus, edge technologies such as AI, especially DL, are applied in cyberspace to check the data collected from physical space, filter out outlier data, make predictions, detect abnormal trends, and prepare good data for applications to use. Various applications are developed in CPSs to realize their assigned functions. The advancements in CPS research will enhance systems' security, scalability, and safety [39].

2.2. Medical Cyber–Physical Systems (MCPSs)

Smart healthcare CPSs aim to monitor patients' health conditions, utilizing the human body as their physical space. The sensitive and dynamic nature of the human body presents several unique challenges. For instance, the IoMT has higher safety requirements [43] since some devices need to attach to a human body [44] to monitor the conditions or detect abnormal situations. Also, the IoMT requires real-time communication, and the system's latency could incur severe results. In the IoMT, the human body moving as a dynamic system also raises some challenges. Thus, designing a reliable, accurate, and fast-response IoMT system in a moving physical space is a big challenge for IoMT systems.

Driven by CPSs, smart hospitals use wearable health monitors [45] and smart beds to gather real-time patient data and share them with healthcare providers for providing interventions and personalized care quickly. The IoMT specifically targets healthcare needs by integrating cyber–physical technologies to enhance capabilities and improve efficiency, safety, and quality of life. However, challenges, including data security [46], interoperability [47], and regulatory compliance, must be addressed for these systems to reach their full potential.

2.3. Internet of Medical Things (IoMT)

With the thriving of the IoT [48], and massive related applications in different areas [49], the IoT shows a lot of potential for benefiting healthcare [50]. The IoMT is the IoT version that only focuses on medical applications [30]. The IoMT defines physical space as any device capable of retrieving data from the human body and any device used to monitor, control, and manage bodily functions. This technology plays an essential role in chronic disease management, remote patient monitoring, telemedicine, and hospital automation, addressing key challenges in healthcare, such as rising costs, staff shortages, and inefficiencies in traditional medical workflows. Wearable health trackers, smart implants, and remote monitoring devices enable continuous tracking of vital signs, facilitating early disease detection and timely interventions. For instance, IoMT-powered ECG monitors can detect irregular heart rhythms in real time, reducing the risk of sudden cardiac arrest. Additionally, smart infusion pumps, connected insulin delivery systems, and AI-driven diagnostic imaging help streamline hospital operations and minimize human errors. The integration of IoMT with AI and DL further enhances predictive analytics, personalized treatment plans, and automated alerts for high-risk patients. With the expansion of 5G networks and edge computing, the IoMT is becoming even more robust, enabling faster

data transmission and reducing latency in critical care applications. As healthcare shifts toward preventive, personalized, and decentralized models, the IoMT serves as the backbone for next-generation smart healthcare systems, enhancing accessibility and quality of care while reducing hospital readmissions and operational costs. The IoMT can improve diagnostic accuracy, enhance surgery operation precision [51], and track patients, vital to further monitoring [52]. In general, the IoMT is a combination of various MCPs to satisfy a specific purpose [34].

2.4. Deep Learning (DL)

DL is a type of AI that uses networks of algorithms called deep neural networks to learn and make decisions from large amounts of data. It works by recognizing patterns in data, such as images or text [53], and improving its accuracy over time. Deep learning is used in many applications, like speech recognition [54], image analysis [55], and healthcare predictions [56]. DL generally has five different approaches, as shown in Table 2. It usually requires five stages: data preparation, model training, model optimization, model evaluation, and model deployment.

Table 2. Different approaches and architectures in DL.

Deep Learning Approach	Subcategories/Description
Supervised Learning [57]	- Classification: Predict discrete labels (e.g., image recognition [58], spam detection [59]). - Regression: Predict continuous values (e.g., predicting house prices [60], stock forecasting [61]).
Unsupervised Learning [62]	- Clustering: Group similar data points (e.g., customer segmentation [63], pattern recognition [64]). - Dimensionality reduction: Reduce data complexity (e.g., PCA [65], t-SNE [66]).
Semi-Supervised Learning [67]	Utilizes a small amount of labeled data with a large amount of unlabeled data to enhance learning performance, such as in text classification.
Reinforcement Learning [68]	- Q-learning [69]: A value-based method to find optimal actions. - Policy gradient [70]: Learning policies directly to maximize rewards.
Federated Learning [71]	A decentralized approach involves training models locally on different devices and then combining them (e.g., distributed learning in mobile networks and privacy-preserving models [72]).

DL can be constructed with different models to deal with different problems. Common DL models include the following:

- Convolutional neural networks (CNNs): A CNN is a specialized deep learning model tailored for analyzing image and visual data. It is widely utilized in medical imaging for tasks such as image classification, segmentation, and anomaly detection, including tumor identification in MRI and CT scans as well as fracture detection in X-rays. Its capability to automatically extract spatial features from images makes it a valuable tool in diagnostic imaging [73]. Unlike traditional rule-based or handcrafted feature extraction methods, which rely on manually defined edge detection filters or statistical techniques, CNNs learn hierarchical representations from raw data, significantly improving accuracy and robustness. Traditional feature extraction approaches, such as edge detection filters (Sobel, Canny) [74] or handcrafted descriptors (HOG, SIFT, LBP) [75], require domain expertise to define relevant features. These methods often struggle with variations in lighting, contrast, and texture, leading to inconsistent results in medical imaging. In contrast, CNNs apply trainable convolutional filters to recognize patterns such as edges, textures, and anatomical structures directly from images, without manual intervention [76]. CNNs employ various convolution techniques [77], such as local connections, weight sharing, and pooling layers, to reduce data dimensions while preserving crucial information. This automation allows CNNs to generalize better across diverse datasets, making them more effective for large-scale, real-world medical applications [78].

- **Recurrent neural networks (RNNs):** An RNN is a specialized artificial neural network for recognizing patterns in sequential data. Examples of such data include time series, speech, or text. Different from traditional feedforward neural networks that process inputs in one direction, RNNs have connections forming directed cycles. This makes them retain a form of ‘memory’ of previous inputs through their internal state. This unique capability makes RNNs highly effective for tasks that require understanding context and sequential information [79]. The RNN is developed to handle sequential data by incorporating loops within the network, allowing information to be carried forward across different time steps. This enables the network to effectively model temporal dependencies, such as those found in natural language processing and financial forecasting [12].
- **Long-short term memory (LSTM):** Introduced by Hochreiter and Schmidhuber in 1997, LSTM networks are an advanced variant of RNNs designed to address the shortcomings of traditional RNNs [80]. Unlike vanilla RNN, which struggles with retaining information over long sequences, LSTM incorporates specialized memory cells designed to preserve information over extended periods. LSTM [81] utilizes three types of gates—input, forget, and output gates—to regulate data flow into and out of these memory cells. This gating mechanism enables LSTM to selectively retain and update information, making it particularly adept at handling tasks that involve long-term dependencies and complex temporal patterns.
- **Gated recurrent unit (GRU):** A GRU [82] streamlines the architecture of LSTM networks by integrating the functions of the forget and input gates into a single update gate. Additionally, GRUs combine the cell and hidden states into one unified representation. This simplification reduces the computing complexity of the model while preserving its capacity to capture long-term dependencies effectively.
- **Bidirectional recurrent neural networks (BRNNs):** A bidirectional RNN [83] enhances a traditional RNN by processing input sequences in both forward and backward directions. This dual-direction approach allows the network to access and utilize past and future context within a sequence. Such bidirectional processing is particularly advantageous in natural language processing (NLP) tasks, where a comprehensive understanding of the entire sentence context is essential for accurate interpretation and analysis.
- **Graph neural networks (GNNs):** Graph neural networks (GNNs) are specialized for analyzing data represented as graph structures, where connections between entities are as crucial as the entities themselves. Unlike traditional neural networks, which process unstructured data, GNNs excel in capturing and analyzing the complex interplay between entities within the graph. They are well suited for applications such as social networks, molecular chemistry, and recommendation systems [84]. The message-passing mechanism [85] in a GNN involves nodes exchanging information with their neighboring nodes. This process allows the network to develop context-aware representations. Graph convolutions [86] extend traditional convolution operations from grid-like data structures to graphs, enabling the network to process nodes while considering the graph’s structure and properties. Attention mechanisms [87] further enhance GNNs by weighing the importance of various neighbors, resulting in more nuanced and context-sensitive representations.
- **Generative adversarial networks (GANs):** Goodfellow et al. [88] described another use of adversarial examples in generative adversarial networks (GANs). Neural networks often learn the probability of training examples (e.g., the probability distribution of the training data). GANs can generate more examples based on the estimated probability distribution. They are based on game theory rather than optimization. Unsupervised

learning using generative models reduces the number of training examples required, the amount of time to train, and the required supervision using training input and output examples. Generative models ignore finding a tractable density function and instead learn a tractable sample generation process in GANs and generative stochastic networks (that came before GANs). Other examples include generative modeling by GANs, variational autoencoders, and fully visible belief nets. GANs are implemented as two ML models playing against each other using game theory and neural networks. One network, the generator, makes samples based on a probability distribution. The other player, the discriminator, examines the samples and makes an estimate of whether they are from the generator (are fake) or from the training distribution (are real).

- **Transformers:** This architecture uses the self-attention mechanism to capture dependencies like eliminating the need for CNN convolutions and RNN recurrence [89]. It is an encoder–decoder architecture that uses the encoder to process the input sequence and the decoder to generate the output sequence. Its positional encodings track the input sequence’s order through recurrence. Transformers have surpassed many state-of-the-art models and have been the foundation of breakthrough models, including Generative Pre-trained Transformer (GPT), Bidirectional Encoder Representation from Transformers (BERT), Text-to-Text Transformers (T5), and Vision Transformer (ViT)—for image processing. It is mainly used in natural language processing (NLP) applications in large language models as well as in various other domains like time-series analysis, computer vision, and reinforcement learning. Their use of multimodal data (e.g., images and text) has enhanced the capability of DL models.

3. Deep Learning in IoMT

DL plays an important role in the IoMT by improving diagnostic accuracy, enabling continuous health monitoring, and supporting personalized treatment plans. For instance, DL models analyze medical imaging data and IoMT sensor outputs to detect diseases, predict health risks, and automate treatment processes like insulin delivery. Additionally, the IoMT supports aging patients, particularly at home, by using smart home sensors to detect intrusion, fire, water leakage, and more, and sending alerts for quick response.

Key technological advancements like edge AI allow lightweight DL models to operate on IoMT devices directly, supporting data processing in real time. Federated learning ensures collaborative training of DL models across devices while preserving patient privacy. Additionally, cloud integration offers scalable storage and advanced analytics for IoMT data, driving the adoption of DL in healthcare systems. The IoMT and DL are transforming healthcare by improving diagnostic precision, enabling real-time interventions, and supporting personalized medicine.

However, this vision of success will come with various new problems. Problems such as standards for data privacy [90], compatibility issues [32], and central processing power in edge devices [91] are a significant obstacle. Furthermore, a critical challenge is having secure and available models [92] that are transferable to various conditions in clinical testing and those being updated.

Healthcare systems face numerous challenges that DL integrated with the IoMT can address, making it a transformative technology for modern medicine. Although the healthcare field continues progressing, issues such as inefficiencies in care delivery, unequal access, increased costs, and suboptimal health outcomes persist. DL in the IoMT solves some social or systemic problems and offers critical contributions to improving efficiency, improving care quality, supporting precision medicine, and enabling groundbreaking research. Table 3 summarizes the benefits and challenges of state-of-the-art applications of DL in healthcare.

Table 3. Application of DL models in IoMT.

Deep Learning Architecture	Application in Medical CPSs	Key Contributions to IoMT	Benefits	Challenges	Example Use Cases
Convolutional Neural Networks (CNNs) [93–95]	Image analysis (e.g., classification, segmentation, detection)	- Highly effective in processing and analyzing 2D–3D medical images. - Task examples include tumor detection, segmentation of organs, and classification of diseases. - Captures spatial hierarchies and features in medical images automatically.	- Highly accurate in detecting and classifying medical images. - Reduces the need for feature extraction manually.	- Large labeled datasets for accurate training. - Expensive, especially for large-scale 3D medical images like MRI and CT scans.	- Cancer detection from mammograms, brain tumor segmentation in MRI. - Retinal image analysis for diabetic retinopathy. - Medical image classification and anomaly detection.
Deep Neural Networks (DNNs) Transformer Models [18]	Multimodal input fusion (e.g., imaging data, clinical data)	- Combine different types of input data to aid in complex diagnostic decision making. - Structured and unstructured data simultaneously. - Prediction.	- Flexible in handling multiple data types. - Complex relationships between clinical and imaging data.	- High risk of overfitting with limited data. - Requires powerful hardware for training deep networks.	- Integrating patient data (genomics, radiomics, clinical data) for personalized treatment. - Decision support for diagnosis and prognosis prediction.
Recurrent Neural Networks (RNNs) [96,97]	Time-series analysis (e.g., vital signs, ECG, EEG data)	- Handles sequential data, making it suitable for physiological data monitoring (e.g., ECG) and continuous real-time patient monitoring. - Useful in speech recognition for medical notes, patient history tracking, and temporal predictions in patient outcomes.	- Effective in analyzing time-dependent data. - Can capture temporal correlations in patient physiological data.	- Struggles with long-term dependencies due to vanishing gradient problems. - Limited performance in handling very long sequences of data.	- ECG and EEG signal analysis. - Predicting the progression of chronic diseases using time-series data. - Real-time vitals monitoring ICUs using sequential patient data.
Long-Short Term Memory (LSTM) [98,99]	Time-series data analysis (e.g., ECG, EEG, blood pressure monitoring)	- Long-range dependencies in time-series medical data (e.g., vital signs, patient monitoring systems). - Predicts trends in physiological data for personalized treatment (e.g., cardiac arrhythmia detection).	- Learning long-term temporal dependencies. - Chronic disease progression modeling based on historical patient data.	- High computational cost compared to GRUs. - Large datasets for training, limiting real-time applications.	- ECG signal analysis for arrhythmia detection. - EEG data analysis for detecting neurological disorders. - Predicting patient deterioration or disease progression in ICUs.
Gated Recurrent Unit (GRU) [100]	Sequential data analysis in real-time systems.	- Similar to LSTMs but with fewer parameters. - Real-time monitoring of patients where rapid feedback loops are necessary. - Effective for sensor data fusion (e.g., integrating data from wearables and imaging devices).	- More efficient than LSTMs. - Real-time systems	- Less effective at capturing very long-term dependencies. - Underperform where intricate, long-range dependencies are critical.	- Real-time analysis of vitals (blood pressure, heart rate) in wearables. - Real-time patient monitoring in ICUs. - Diabetes management, blood glucose levels for insulin pump.

Table 3. Cont.

Deep Learning Architecture	Application in Medical CPS	Key Contributions to IoMT	Benefits	Challenges	Example Use Cases
Bidirectional Recurrent Neural Networks (BRNNs) [101]	Sequence learning for tasks requiring context from both past and future	- Processes input sequences in both directions, capturing complete temporal context. - Used in natural language processing (NLP) applications in healthcare. - Useful for analyzing sequential medical data (e.g., drug response prediction, medical transcription).	- Captures more contextual information from both directions. - Full context is essential, such as text analysis in medical records.	- Requires more computational resources than unidirectional RNNs. - Not suitable for real-time applications due to backward pass requirements.	- Medical note analysis and automated report generation. - Predicting patient responses to treatments based on complete historical medical records. - Drug interaction analysis based on patient history.
Graph Neural Networks (GNNs) [102]	Modeling relationships between entities in medical data	- Effective for modeling structured, relational data in medical contexts (e.g., modeling protein interactions, and disease pathways). - Used for analyzing connections between patients, diseases, or drugs in healthcare systems. - Tasks like patient similarity modeling, social determinants of health, and personalized medicine.	- Captures complex relationships between medical entities. - Personalized treatment recommendations based on patient relationships.	- High computational complexity for large graphs. - Requires well-defined, structured data, which is not always available in medical contexts.	- Predicting drug-drug interactions based on patient histories and molecular graphs. - Disease comorbidity analysis in patient networks. - Protein structure prediction in bioinformatics.
Generative Adversarial Networks (GANs) [103]	Data augmentation, synthetic image generation	- Can generate synthetic medical images that can be used for data augmentation in training deep learning models, especially when limited data are available. - Generating high-quality medical images that closely resemble real data, improving model robustness. - Enhancing low-quality medical images or filling in missing data.	- Helps address data scarcity issues, improving model accuracy. - Provides diverse training data for robust model generalization.	- Adversarial training is often unstable, leading to mode collapse. - GAN-generated images can introduce bias if not carefully validated.	- Creating synthetic MRI, CT, or X-ray images for rare diseases. - Enhancing noisy or low-resolution medical images. - Generating missing regions in medical images.

3.1. IoMT Architecture

After we review different frameworks and architectures from different sources [104–106], we improvise and propose our architecture of DL-empowered IoMT for healthcare, as shown in Figure 1. The whole structure consists of four layers as follows:

- Service layer: This layer is the headquarters of the system; it can be used as data storage or for sending requests to the user-end and training as a global model aggregator in deploying federated learning [107].
- Network layer: This layer serves as a reliable carrier, making sure to smoothly transmit information to the end-user. It is adaptable to various communication technologies such as LTE [108], 5G [109], and WiFi [110], making it a practical and versatile component of the architecture.
- Edge layer: This layer is an edge remote server layer; it can provide computing resources for the end-users.
- Smart device layer: This layer, comprising a variety of IoMT devices, is designed to be flexible and adaptable. It can sense, send, or collect patient data, and can be deployed in both local and centralized model training, ensuring its versatility and adaptability.

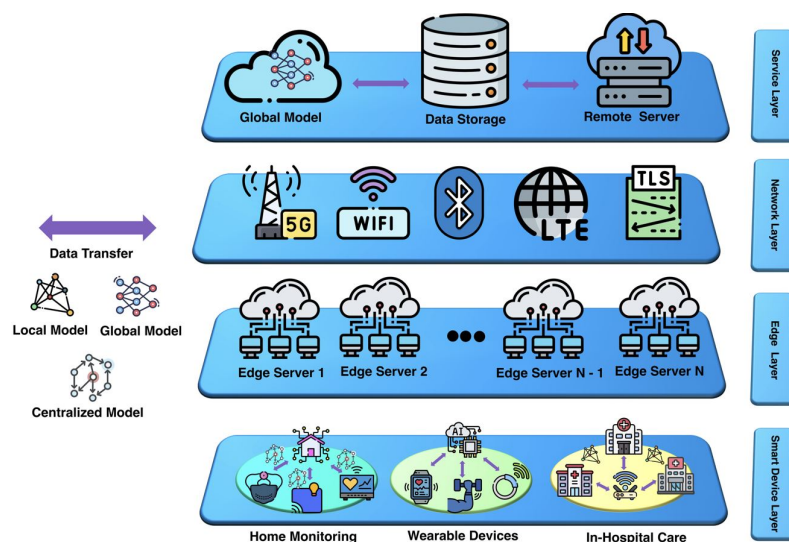


Figure 1. Layer architecture of smart healthcare.

As shown in Figure 1, the architecture of the IoMT comprises several key components:

- Sensors: These are at the core of the IoMT and are embedded in places (e.g., wearable devices, smart implants, or ambient monitoring systems). Sensors capture physiological signals like heart rate, glucose levels, blood pressure, oxygen saturation, or movement data.
- Communication networks: The IoMT relies on secure and robust communication technologies like Bluetooth, Wi-Fi, Zigbee, and 5G to transfer data from sensors to processing systems. These networks must ensure low latency and high reliability for supporting real-time healthcare applications.
- Edge devices: Often situated close to the sensors, edge devices preprocess data, reducing latency and offloading computation from the cloud. For instance, smartwatches or edge processors can analyze sensor data locally to detect anomalies or trigger alerts.
- Cloud platforms: Cloud computing provides scalable infrastructure to store and analyze the large amounts of data generated by the IoMT. Cloud platforms support advanced data analytics, integration of multimodal data, and collaborative healthcare services, among others.

3.2. Information Synthesis in IoMT

The exponential growth in patient-generated medical data presents challenges in managing, interpreting, and using this information. DL-powered IoMT systems offer robust solutions for synthesizing and analyzing this data.

- **Patient data:** IoMT devices continuously generate and collect massive amounts of data in real time, including vital signs, activity levels, and chronic condition metrics. DL models can integrate these datasets with clinical records to provide actionable insights for personalized care.
- **Data complexity:** IoMT devices produce heterogeneous and high-dimensional data, such as time-series signals from wearables, imaging data from portable X-rays, and multimodal data streams from smart hospital systems. DL excels at processing such complex data and uncovering patterns that are otherwise challenging to identify.
- **Medical literature integration:** With more than a million citations added annually to the medical literature, DL models trained on vast textual datasets can synthesize knowledge, helping healthcare professionals stay up to date with the latest clinical advancements. According to AI pioneer Geoff Hinton, DL would provide better results in medical science than a radiologist [111].

3.3. Augmenting Human Performance

The increasing complexity of patient presentations and treatment regimens requires tools that enhance human decision making. DL integrated with the IoMT can improve clinicians' ability to manage and interpret medical information effectively.

Experts' experience, knowledge, or other factors can heavily affect the results of the medical image interpretation. Machine learning (ML) models can assist experts to some extent, but the models may lack flexibility when faced with vast amounts of data with complex problems [112]. Convolutional neural networks (CNNs) in DL have shown exceptional performance in image classification tasks, such as identifying tumors in MRI scans or detecting fractures in X-rays [113]. Based on a systematical review from A Fourcade and RH Khonsari, a CNN is not a replacement for doctors but an optimization of routine medical tasks [114].

DL can be utilized to improve efficiency in the following areas:

- **Rare and complex cases:** DL models trained on diverse datasets are capable of providing decision support for rare diseases and atypical presentations, reducing diagnostic errors and delays. IoMT-enabled devices collect real-time data that DL algorithms can analyze to identify anomalies and suggest interventions [112–114].
- **Multimorbidity management:** The increasing prevalence of patients with multiple comorbidities complicates diagnosis and treatment. IoMT systems combined with DL can identify interaction effects and recommend optimized treatment pathways.
- **Skill augmentation:** DL-powered IoMT solutions can democratize knowledge by providing diagnostic and therapeutic recommendations informed by vast datasets, reducing disparities in care quality. For instance, Groh et al. analyzed the diagnosis of skin diseases on dark skin. Without DL, the accuracy rate of board-certified dermatologists (BCDs) was 37%. Once DL was added, the accuracy rate of the BCDs was improved to 60%. The same result was applied to primary care physicians (PCPs), whose accuracy rate improved from 17% to 47% [115].

DL improves almost every aspect of healthcare [116], such as its performance becoming more accurate [117] and the system being more robust [115]. With the framework being more secure, we intend to focus on studying and deploying different applications of deep

learning in healthcare to enhance its performance based on the unique requirements of medical CPSs.

4. Problem Space for DL Applications

To better examine the performance of DL applications in the IoMT, we propose three orthogonal dimensions of the problem space, as shown in Figure 2. It contains three dimensions. (i) Device deployments: This concerns how the IoMT is to be deployed, and it can be in home devices, wearables, or just in the hospital. (ii) Architecture layers: This is from the IoMT architecture angle based on the framework in Section 3.1 to examine which layers (sensor, network, and application) are enhanced by DL. (iii) Application scenarios: Based on different patient requirements, each scenario needs the IoMT to be optimized.

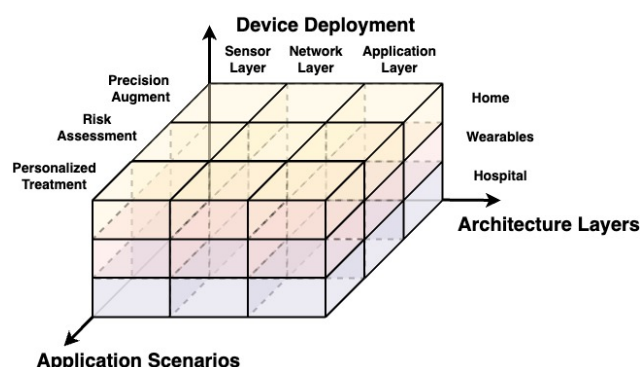


Figure 2. Problem space for DL-empowered IoMT.

We summarize the application scenarios into three representative key application scenarios. (i) Precision augmentation: This is about enhancing the accuracy of diagnosis and clearance of medical operations. (ii) Risk assessment: This is about predicting the chances of risks for postoperative recurrence or the likelihood of some potential complications after surgery to improve success in medical treatment. (iii) Personalized treatment: This is the customization of medical treatment and daily vital data for analysis to enhance quality of life and mental health.

4.1. IoMT Architecture Layers

The IoMT is a multi-layered system designed to enhance healthcare by integrating medical devices, sensors, networks, and AI to support real-time data collection, processing, and decision making. The conceptual framework for the IoMT consists of several key components, each playing a crucial role in ensuring efficient healthcare delivery. These components interact in a structured manner to enable seamless data exchange, remote monitoring, predictive analytics, and automated interventions. The IoMT framework consists of several layers that are detailed below.

4.1.1. Smart Device Layer

The smart device layer forms the foundation of the IoMT by capturing real-time medical data from diverse sources, enabling continuous monitoring, early disease detection, and timely interventions. It includes a wide range of sensing devices, such as wearable sensors, implantable medical devices, smart hospital equipment, remote monitoring systems, and environmental sensors, all designed to collect vital physiological parameters and behavioral data. Wearable devices like smartwatches, ECG monitors, and glucose sensors continuously track health related status like heart rate, blood pressure, oxygen saturation, and glucose levels, offering insights into a patient's health trends. Implantable medical devices, such as pacemakers and neurostimulators, monitor and regulate critical bodily functions,

enhancing the quality of life for patients with cardiovascular and neurological disorders. Remote monitoring systems, including blood pressure monitors, spirometers, and respiratory sensors, enable patients with chronic conditions to be monitored outside hospital settings, reducing the need for frequent hospital visits and minimizing healthcare costs.

Also, smart hospital equipment, including automated infusion pumps and AI-powered drug dispensers, ensures precise medication administration while enhancing patient safety and operational efficiency. Environmental sensors that track temperature, humidity, and air quality play a crucial role in preventing hospital-acquired infections (HAIs) and ensuring a safe clinical environment, especially for ICU and immunocompromised patients. Interaction within the IoMT ecosystem: Once the smart device layer collects data, secure communication protocols (e.g., Bluetooth Low Energy (BLE), Zigbee, Wi-Fi, 5G, LoRaWAN) facilitate real-time transmission to network and computing layers for further analysis.

To minimize latency and bandwidth overload, edge AI models deployed on local devices can perform preliminary anomaly detection, noise reduction, and redundancy filtering before sending critical insights to central servers. For example, an AI-enabled ECG monitor can detect abnormal heart rhythms, such as atrial fibrillation, and trigger an alert for immediate intervention. Additionally, federated learning can be used to train AI models across multiple IoMT devices without transferring raw patient data, enhancing privacy and security while improving diagnostic accuracy. By integrating IoMT sensing devices with AI-driven edge computing, the smart device layer ensures high-quality medical data acquisition, paving the way for personalized, real time, and proactive healthcare solutions. Table 4 lists some studies focused on sensors attached to different devices with DL to satisfy specific requirements.

Table 4. Related works in sensor layer.

Category	Home	Wearables (Including Sensor Information)	Hospital
Precision Augmentation	Parkinson's disease evaluation [118].	Atrial fibrillation early detection with smartwatches [119], or device STAT-ON. The sensor layer includes sensors such as accelerometers to get position on the body [118].	Detection of atrial fibrillation [120].
	Heart disease remote monitoring system [121].	Heart monitoring [122]. The sensor layer includes sensors such as blood pressure, oxygen, and heart rate [121].	Surgical robotic systems [7,123–125].
	Stress-level detection [126,127].	Three stress tests were conducted with a wearable system that relies on electrocardiogram (ECG), respiration, skin conductance, as well as surface electrocardiogram (sEMG) [126].	Fuzzy logic DL in iStree application [126].
		Stress-track-sensing wrist band with sensors to measure temperature, humidity, and steps [127].	Trained a meta-model on stress prediction values to improve the model's performance [127].
Risk Assessment	Healthcare Monitoring System (i-NXGeVita) [128].	ECG sensor (module ad8232) collects heartbeats (heart signals) of the patient [128].	i-NXGeVita (TensorFlow + GPU to train the model) [128].
		Risk factor monitoring such as stroke detection with sensors to test oxygen [129], ECG [128], etc.	Smart surgery [130], smart laparoscopic forceps in minimally invasive surgery system [131].
Personalized Treatment	Stress recognition in smart home [126,127,132].	Biomedical systems for intelligent healthcare [29,126].	Health monitoring in diabetics [133], or fuzzy logic DL in iStree application [126].
	Prediction of sleep stages [134].	Self-management of cardiovascular conditions [135].	Wheelchair control enhancement [123].

4.1.2. Network Layer

Adopting IoMT systems requires a large volume of medical data to be passed through network systems, including high-resolution medical images, real-time monitoring data, and

patient records. Although current network systems have significantly improved bandwidth, the exponential growth in IoMT devices and the associated data creates potential risks such as data overflow, network congestion, and latency. These challenges could compromise the reliability, accuracy, and timeliness of medical diagnostics and treatment delivery [136]. DL can be leveraged to optimize the performance of IoMT systems by reducing the data load on network infrastructures while maintaining the accuracy required for medical decision making. For instance, federated learning allows processing and analysis of medical data—such as for images, sensor readings, or real-time monitoring streams locally or at the edge. DL models can extract essential features, compress data, or perform preliminary diagnostic assessments before sending data to central servers. These methods minimize the amount of raw data transmitted through the network, ensuring efficient bandwidth utilization while preserving diagnostic accuracy to enable healthcare professionals to make reliable and informed decisions.

The network layer is responsible for bridging the perception layer (sensing devices) and the AI-powered decision-making layer. Data from wearable and implantable IoMT devices are transmitted securely via wireless communication protocols, processed at edge nodes for immediate insights, and then sent to cloud-based platforms for further AI analysis. Healthcare providers receive real-time notifications and alerts, allowing them to intervene proactively, manage chronic diseases efficiently, and reduce hospital readmissions. By leveraging advanced wireless technologies, distributed AI models, and blockchain security, the network layer ensures seamless, efficient, and privacy-preserving healthcare communication, paving the way for next-generation smart healthcare ecosystems.

To enhance network efficiency in IoMT systems, DL can be applied to reduce redundant data transmission, optimize bandwidth use, and improve timely data processing. Federated learning allows IoMT devices to process data locally, instead of continuously transmitting raw data to centralized servers. This way can reduce network congestion and latency while maintaining diagnostic accuracy [136]. Additionally, DL-based data compression techniques can encode high-resolution medical images and real-time monitoring data in a more compact format without compromising critical diagnostic features. Edge computing integrated with DL further optimizes data flow by preprocessing, filtering out redundant information, and only transmitting essential insights [91]. This can not only reduce bandwidth consumption, but also lead to faster decision making and real-time responsiveness in critical care scenarios. Anomaly detection models powered by DL can analyze data streams in real time, identifying and discarding outliers or duplicated information before transmission, minimizing unnecessary network load. Additional research should develop adaptive DL models that dynamically adjust data transmission rates based on network conditions, further enhancing the scalability and reliability of IoMT systems.

Table 5 indicates related works concentrating on the DL in network infrastructures.

Table 5. Related works in network layer.

Category	Home	Wearables	Hospital
Precision Augmentation	Dementia analysis [137].	Cardiovascular monitoring for COVID-19 patients [138].	Disease screening and enhancement of smart healthcare [139].
	Cloud-based biomedical applications [140].	Deep belief network for heart disease diagnosis [141].	Reinforcement learning for healthcare workflows [142].
Risk Assessment		Tracking vital signs of patients [143].	Remote robotic surgery [144].
Personalized Treatment	Smart Dental Health–IoT platform [145].	Elderly healthcare framework [146,147].	Smart hospital operations [147,148].
	Remote patient monitoring systems [149].	Deep learning framework for data analytics [150].	AI-IoMT systems for disease prognosis [151].

4.1.3. Application Layer

DL in the application layer supports creating personalized treatment plans via analyzing individual patient data such as genetic, demographic, and clinical history. This layer enhances clinical workflows, patient engagement, and telemedicine services through AI-driven applications. Clinical decision support systems (CDSSs) assist healthcare professionals by analyzing patient data for accurate diagnostics, risk assessments, and treatment recommendations. AI-powered CDSSs can detect early signs of diseases and suggest optimal intervention strategies [152]. Remote patient monitoring (RPM) facilitates continuous home-based care, reducing hospital visits while ensuring timely interventions for patients with chronic conditions like hypertension, diabetes, and heart disease [153]. Telemedicine platforms facilitate virtual consultations, allowing real-time interaction between doctors and patients.

Also, AI-driven chatbots and virtual assistants further support triage, symptom analysis, and medication adherence. Personalized treatment systems use AI to facilitate treatment plans, medication dosages, and lifestyle recommendations according to individual patient data for improved outcomes. Interaction in the IoMT ecosystem: Healthcare providers receive real-time notifications and reports via mobile and web-based dashboards, improving decision making and response times. Patients interact with AI-powered apps, improving engagement, treatment adherence, and self-management of health conditions. By integrating AI-driven analytics, real-time alerts, and patient engagement tools, the application layer bridges the gap between IoMT-generated data and actionable healthcare decisions, enhancing efficiency, accessibility, and personalized patient care. Table 6 lists some examples of applications with DL to satisfy personalized treatment.

Table 6. Related works in application layer.

Category	Home	Wearables	Hospital
Precision Augmentation	Tracking movement disorders [154].	Respiratory sound classification [155].	Medical diagnostics enhancement [7,156,157].
	Smart bed [94,95,158–160].	Cardiovascular disease detection [161].	Improved cancer detection [12,162].
Risk Assessment		Predicting postoperative complications [163].	Predicting serious complications [164].
Personalized Treatment	IoMT-based smart healthcare of elderly people [165].	Multi-disease detection continual learning framework [166].	Parkinson’s disease [157,167].
	IoMT framework for home healthcare [168].	Pervasive diabetes diagnosis [169].	Heart failure and arrhythmia detection [170].

4.2. Device Deployment with DL

The “device deployment” axis aims to identify and analyze the locations where the IoMT technologies will be implemented.

4.2.1. DL Applied in Home Devices

Some devices are designed to be installed in a patient’s home to monitor daily activities and detect potential risks, such as falls or irregular movements. These devices typically utilize Local Access Network (LAN) technology such as WiFi or Bluetooth (BT) to transmit data over relatively short distances. Moreover, they are often wearable and powered by batteries to ensure portability and ease of attachment to the patient’s body, enhancing convenience and usability in a home setting. Some devices are installed within the house to monitor the environment, such as room temperature and humidity, to ensure a proper room environment suitable for patients.

Using DL with the IoMT for Parkinson’s disease is an example. DL offers several advantages because it handles complex, high-dimensional data and extracts meaningful patterns. Parkinson’s disease symptoms, such as tremors, rigidity, and bradykinesia, often appear late

in disease progression. DL models can analyze subtle patterns in medical imaging or wearable sensor data, enabling earlier detection. A study [167] designed a DL-based FPCIT SPECT interpretation system to enhance understanding of Parkinson's disease. The work showed that this approach can mitigate interobserver variability. Afterward, in 2019, DL techniques were utilized to classify PD using a nine-layered CNN on vocal (speech) features [157]. The study provided compelling evidence of the classifier's efficacy in addressing this challenge.

Elderly fall is another common concern for elderly persons. Within 2014, there were 28.7% elderly falls, 7 million injuries, 27,000 elderly persons died, and 800,000 were hospitalized because of falling [171]. DL algorithms, such as LSTM, installed on home cameras can be used to detect and predict elderly falls according to the falling position [172,173] such as forward laying, front knees, sideward laying, and back sitting [173]. Timely response with DL applications can increase the chances of quick treatment following a fall.

4.2.2. DL Applied in Wearable Devices

The value of the global wearable technology market has been projected to grow dramatically. The value increase from USD157.30 billion in 2024 to USD1695.46 billion by 2032. Active wearables include smart glasses, VR headsets, and smartwatches. Passive wearables include wristbands, rings, key rings, brooches, and clothing [174].

DL applications in healthcare monitoring include DL use in wearable devices with sensors for tracking patient vitals and movements to enhance care [122]. The rise of wearable technology, such as smartwatches and ECG monitors, is driven by an aging population. These devices collect substantial health data that can be analyzed with DL for condition monitoring and prediction. Studies often employ CNNs [175] to process ECG signals for arrhythmia detection and RNNs to analyze heart rate data for predicting cardiac events.

DL models, especially CNNs, embedded in wearable devices to predict and record activities (e.g., sitting, walking, lying down, climbing stairs, jogging, running, falling) can achieve human activity recognition (HAR). CNNs are widely used in image recognition, with excellent short-distance signal processing [176].

4.2.3. DL Applied in Hospital Devices

Using DL in hospitals provides transformative potential to enhance patient care, improve operational efficiency, and advance medical research. DL with the IoMT has been applied in the following aspects.

- **Smart bed:** A smart bed is a piece of medical equipment prominently applied in intensive care for monitoring bed-bound patients as well as collecting data for diagnosis [159] and predictions [94]. A CNN [158] is integrated with a long short-term memory model to enhance performance relative to previous models to prevent terminal patients developing pressure ulcers. Research has shown the application of CNNs [95,160] in processing sensor data for forecasting patient outcomes and preventing bedsores.
- **Robotic surgery systems:** Robotic surgery systems, like the Da Vinci Surgical System, use DL for improved precision and control. CNNs analyze visual data from surgical cameras, aiding surgeons with tissue recognition and instrument tracking. RNNs help predict surgical tool movement for safer operations. DL has made significant strides in autonomous systems and robotics, with applications ranging from self-driving cars to robotic process automation. In robotics, DL [7] has enhanced object recognition, manipulation, and navigation capabilities, allowing robots to perform tasks with greater precision and adaptability [177].

We comprehensively reviewed medical devices equipped with DL technologies, as summarized in Table 7.

Table 7. Overview of medical devices in IoMT with DL models [2].

Device/Sensors	Use of Device	Benefits	Drawbacks	DL Integration	DL Model
MicroOptical™, Westwood, MA, USA [178]	Orthopaedic fluoroscopy	Image quality is good, low radiation exposure	Lag time, difficulties in focusing	DL algorithms can reduce lag time, enhance image focus, and improve real-time visualization in fluoroscopy procedures.	CNNs, GANs, etc.
Opti-Vu HDVD, Wayzata, MN, USA [179]	Laparoscopy HMD vs. video monitor	Smooth motion	Imbalance weight, single goggle operation	DL-powered motion tracking can improve stability, while real-time image enhancement can improve surgical precision.	CNNs, RNNs, GANs, etc.
ANC CMOS and Google GLASS, CA, USA [180]	Image-assisted cancer resection surgery	Lightweight, wearable, less interference with surgery	Costly, lag time, bad mobility, need wireless data transport	DL-based image segmentation can improve tumor detection and surgical accuracy.	CNNs, GANs, etc.
Richard Wolf GmbH, Knittlingen, Germany [181]	Trans-anal endoscopic microsurgery	Comfortable with peripheral view, easy to use	Poor image quality	DL models can enhance image resolution and clarity for better surgical outcomes.	CNNs, GANs, etc.
Nomad ND2000™, Stafford TX, USA [182]	Displays vital signs monitoring for general anesthesia	Quick response	Good for general condition	DL algorithms can predict anomalies in vital signs and automate alerts for abnormal conditions.	CNNs, RNNs, GANs, etc.
HMZ-T2 and Wrap1200, Sony Corporation, Tokyo, Japan [183]	Sonography to educate patient	Compact, clear disease visualization, high imaging quality	Lag time, power limitation	DL can reduce lag time and enhance sonography image interpretation for better education and diagnostics.	CNNs, GANs, etc.
Google GLASS, CA, USA [184]	Transformation forensic medicine	Easy use, wearable	Bad-quality images	DL models can enhance image quality and automate forensic analysis for efficiency.	CNNs, GANs, etc.
Primesense 1080/Xtion, Tel Aviv, Israel [185]	Assistive navigation for visually impaired patients	Easy use, provides depth and distance	No studies conducted	DL-powered depth estimation and object detection can improve navigation accuracy for visually impaired patients.	YOLO, R-CNNs, etc.

Table 7. Cont.

Device/Sensors	Use of Device	Benefits	Drawbacks	DL Integration	DL Model
ViSi Mobile System, USA [186]	Vital signs monitoring	ECG, RR, HR, SpO2, pulse rate, skin temperature measured, less costly	Not enough results	DL algorithms can monitor vital signs in real-time to identify anomalies and predict potential cardiac risks.	CNNs, RNNs, GANs, etc.
Aingeal, Ireland [187]	Monitor vital signs	Device is small and patients preferred it, lower false positive rate, automated	Not enough results	DL models can improve arrhythmia detection accuracy and reduce false negatives in patient monitoring.	CNNs, RNNs, GANs, etc.
QUASAR [188]	ECG monitor	ECG results are clear, small in size	P-waves are not viable	DL algorithms can reconstruct missing P-waves and enhance ECG signal quality for better cardiac diagnostics.	CNNs, RNNs, GANs, etc.
BioHarness, USA [189]	Monitors heart and respiratory rate, activity and posture	Multiple clinical trials have demonstrated high sensitivity and specificity with positive outcomes	Costly	DL can provide predictive analytics for health monitoring, including fall detection and activity tracking.	GNNs, RNNs, etc.

4.3. Application Scenarios with DL

Using DL in medical applications offers groundbreaking potential to improve health-care delivery, enhance patient outcomes, and support medical research. Here are the key reasons why DL is increasingly being adopted in medical applications.

4.3.1. Precision Augmentation

Medical imaging is one of the most prominent areas where DL, especially CNNs, has been extensively applied. CNNs have shown exceptional performance in image classification tasks, for example, identifying tumors in MRI scans or detecting fractures in X-rays [113,190]. A notable study applied a deep CNN to classify breast cancer images, achieving higher accuracy than traditional methods. The study highlights the critical applications of DL, particularly in smart healthcare systems, providing examples like medical imaging, diagnostics, and patient monitoring systems [12]. It identifies and discusses emerging research trends, including federated learning, explainable AI, and the integration of DL with edge computing. Existing applications are discussed below.

- **Multiple sclerosis:** Latent patterns of multiple sclerosis (MS) lesions were identified from initial images using a CNN methodology [191]. Another study [192] proposed employing a DBN and random forest (RF) in conjunction with myelin and T1-weighted images to recognize MS pathology in brain tissue, which showed as normal on an MRI. They utilized a four-layer DBN to learn latent feature representations from 3D image patches of normal-appearing white matter (NAWM) and normal-appearing gray matter (NAGM) in this approach [192].
- **Cancer detection:** In medical diagnostics, deep learning has shown significant potential to revolutionize healthcare. One study explored a computer-aided diagnosis system that utilizes deep learning algorithms for detecting, segmenting, and classifying masses in mammograms [162]. It explored several deep learning methods for mass detection, including CNN, Deep Belief Network (DBN), and Conditional Random Fields (CRFs). Previous studies [193–195] have shown that classical learning techniques (SVM, RF, decision tree) perform well in terms of predicting breast cancer survival.
- **Heart disease detection:** Some studies have demonstrated that CNNs can detect heart failure and arrhythmia from image collections obtained via IoT devices analyzing ECG data [170]. They facilitated faster training and testing processes by categorizing ECG data into two distinct heartbeat signals. Their simulation evaluated the CNN's efficacy with ECG image datasets, utilizing cloud resources for swift and lag-free classification. The results showed an improved accuracy rate, surpassing existing ECG image classification algorithms and highlighting CNN's superior performance in identifying instances of heart conditions [196].
- **Alzheimer's disease detection:** Some studies [197] proposed a CNN-driven method for conducting early-stage Alzheimer's disease (AD) diagnosis using structural MRI [198]. A study [199] developed a CNN-based procedure to diagnose AD from structural MRI, demonstrating superior performance compared to SVM in their study. Ref. [200] recommended an RNN-based model to forecast the progression from mild cognitive impairment (MCI) to AD, extracting temporal variables from multimodal data.

4.3.2. Risk Assessment

A study presented an innovative computer-aided diagnosis system for epilepsy, utilizing CNN to analyze encephalogram data. It evaluates the performance of a 13-layer CNN model by comparing its specificity, accuracy, and sensitivity with other machine learning

techniques in detecting seizures across normal and preictal classes [156]. This represents a significant application of deep learning in epilepsy diagnosis.

The emerging use cases of IoMT in healthcare span a wide range of transformative applications, including real-time health monitoring, remote patient care, and predictive analytics. These innovations enable continuous tracking of vital signs, facilitate home-based disease management, and anticipate health events like heart attacks. The IoMT also enhances surgical precision through smart systems, supports virtual consultations via telemedicine, and improves medication adherence with smart devices. Additionally, the IoMT plays a critical role in care for the elderly, personalized medicine, mental health monitoring, and diagnostic imaging while optimizing hospital operations, rehabilitation, and resource management. These use cases collectively improve healthcare accessibility, efficiency, and patient outcomes. Some of the most promising and emerging use cases of the IoMT in healthcare are presented in Table 8.

Table 8. IoMT use cases in healthcare.

Use Case	Description	Example	Impact
Real-time health monitoring	IoMT devices support continuous monitoring of vital signs like heart rate, blood pressure, and oxygen saturation	Smartwatches track heart rate and detect anomalies.	Improved chronic disease management and real-time health insights.
Remote patient monitoring (RPM)	Devices monitor patients remotely, reducing hospital visits and enabling home-based care.	IoMT devices managing chronic diseases like diabetes remotely.	Reduced hospital burden and enhanced rural healthcare accessibility.
Predictive healthcare analytics	IoMT data combined with AI predict health events and outcomes like heart attacks or hospital readmission.	Platforms predicting sepsis in ICU patients using IoMT data.	Better preventive care and fewer emergency interventions.
Smart surgical systems	IoMT-enabled robotic surgical systems enhance precision and reduce human error during surgeries.	Da Vinci Surgical System using IoMT for minimally invasive surgeries.	Increased accuracy and safety in complex surgical procedures.
Virtual healthcare (telemedicine)	Connected devices facilitate remote consultations and diagnostics through telemedicine platforms.	Connected stethoscopes and diagnostic kits for virtual consultations.	Broader access to healthcare, especially in remote regions.
Medication adherence monitoring	IoMT systems track medication usage with smart devices like pillboxes or connected dispensers.	Smart inhalers tracking usage and sending reminders.	Enhanced patient compliance and reduced treatment failures.
Assisted living and elderly care	IoMT devices like fall detectors and trackers improve safety and independence for the elderly.	Wearable fall detectors alerting caregivers in emergencies.	Improved safety and quality of life for aging populations.
Personalized medicine	The IoMT collects data to deliver tailored treatment plans based on health metrics, lifestyle, and genetics.	Continuous glucose monitors adjusting insulin dosages automatically.	Precise treatments with reduced side effects.
IoMT in mental health	Wearables and apps monitor indicators like stress and sleep patterns to detect mental health issues.	Smartwatches tracking sleep disturbances for early mental health intervention.	Proactive mental health management and early intervention.
Connected medical imaging	IoMT-enabled imaging devices provide real-time transmission of images for faster diagnosis.	Portable X-ray devices uploading images for real-time analysis.	Faster and more accurate diagnostic imaging in critical cases.

Table 8. Cont.

Use Case	Description	Example	Impact
Smart hospitals	IoMT systems optimize workflows and patient safety using connected devices and analytics.	Smart beds detecting falls and monitoring patient vitals.	Increased operational efficiency and patient safety.
IoMT in rehabilitation	Smart prosthetics and sensors assist in rehabilitation and progress tracking for patients.	AI-powered prosthetics adapting to movement patterns.	Enhanced rehabilitation outcomes and independence for patients.
Supply chain and asset management	IoMT tracks medical equipment and supplies for better availability and maintenance.	Trackers monitoring the location of ventilators and other equipment.	Improved resource allocation and reduced equipment downtime.
Infection control	IoMT devices monitor hygiene and sterilization practices to reduce infection risks.	Connected devices tracking hand hygiene compliance in hospitals.	Lower infection rates and enhanced hygiene compliance.

4.3.3. Personalized Treatment

The integration of DL into the IoMT has created massive healthcare applications, as shown in Figure 3. Hence, we summarize a few different DL applications in healthcare with IoMT.

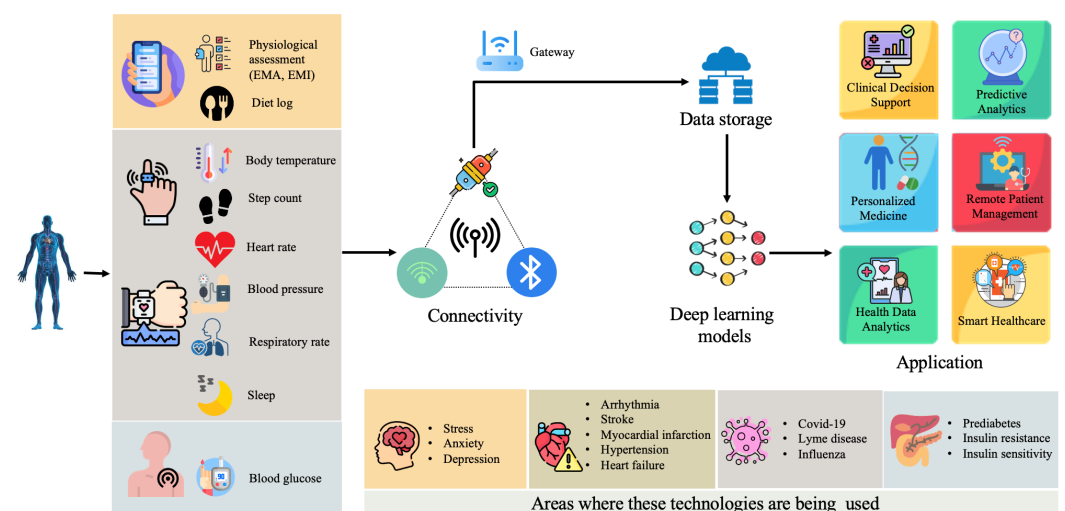


Figure 3. Workflow of DL models in IoMT for healthcare.

4.3.4. Other Applications

There are additional applications of DL in *medical asset tracking and hospital operations*. Efficient medical asset tracking ensures that critical equipment is available when needed, reducing delays in patient care. One study [37] reviews advancements in predictive maintenance (PdM) strategies that anticipate equipment failures to optimize maintenance and reduce downtime. It examines various PdM methodologies by system, purpose, and approach, highlighting their effectiveness. Another study also explores PdM systems, their purposes, and effective maintenance strategies in smart homes. It categorizes PdM approaches, discusses their roles, and examines how AI, machine learning, and IoT enhance predictive maintenance systems [201].

4.3.5. Application Performance

Table 9 highlights key case studies and best practices of DL applications in the IoMT, showcasing their transformative impact on healthcare. It covers innovative implementations in areas such as remote monitoring, personalized medicine, diagnostic imaging, and

hospital resource optimization. By examining real-world examples, this table underscores the potential of the IoMT and DL to improve healthcare delivery and enhance patient outcomes, as well as lead to efficient operations.

Table 9. DL in IoMT: Case studies and best practice.

Case Study	Real-Life Example	Remarks
Remote patient monitoring for chronic diseases	Apple Watch detecting atrial fibrillation using ECG data.	Robust preprocessing, battery efficiency, and user adoption are critical.
Predictive analytics in smart hospitals	Mount Sinai Hospital’s DL-powered system reducing sepsis mortality.	Interoperability, low-latency networks, and diverse training data improve results.
Medical imaging and IoMT integration	CoviDetect X-ray scanners aiding rapid COVID-19 diagnosis.	Data augmentation and cloud integration enable rapid, accurate diagnostics.
Personalized medicine	Medtronic’s AI-powered insulin pumps for diabetes management.	Privacy-preserving techniques and adaptability enhance personalized care.
Smart beds for patient monitoring	IoMT-enabled smart beds reducing fall injuries in elderly care.	Reliable sensors and edge AI improve real-time monitoring accuracy.
Early detection of neurological disorders	DL models analyzing EEG and sensor data for early Parkinson’s diagnosis.	Feature engineering and longitudinal data improve diagnostic reliability.
Assistive technologies for elderly and disabled patients	Ossur’s AI-powered prosthetics adapting to user movements.	Human-centric design and robust continuous monitoring ensure usability.

Table 10 presents a comparative analysis of various DL models applied in different IoMT domains. The table highlights the applications of these models, along with a performance comparison between traditional technologies and DL-enhanced approaches. To ensure a straightforward and quantifiable comparison, we primarily focus on accuracy rate as the key performance metric. For instance, the researchers in [202] deployed a CNN-based ResNet-50 model to assist in detecting COVID-19 from CT scans. The model was trained using a combination of the COVID-CT dataset and proprietary hospital data, achieving 97% accuracy with a 55 ms inference time. To enhance robustness, data augmentation and transfer learning were applied, improving model generalization. This case study highlights the critical role of CNNs in rapid and accurate COVID-19 detection in IoMT applications [202].

Table 10. DL performance.

DL Model	Application in IoMT	Traditional Technology Performance	Enhanced Performance with DL
Convolutional Neural Networks (CNNs) [93–95]	Image analysis (e.g., classification, segmentation, detection)	70% accuracy, manual feature extraction [203]	Accuracy: COVID-19 CT image detection: 97.0% with VGG19 algorithm [202]
Deep Neural Network (DNN) Transformer Models [18]	Multimodal input fusion (e.g., imaging data, clinical data)	With machine learning (ML) models [204]	Grows in accuracy by 15% and reduces in time complexity by 32% [204]
Recurrent Neural Networks (RNNs) [96,97]	Time-series analysis (e.g., vital signs, ECG, EEG data)	High false alarm rate, difficulty detecting complex patterns	90%+ accuracy in anomaly detection, real-time alerts [205]
Long-Short Term Memory (LSTM) [98,99]	Time-series data analysis (e.g., ECG, EEG, blood pressure monitoring)	Accuracy of 76.6% with Wind-Driven Optimization (WDO) [206]	Accuracy of 97.6% [206]

Table 10. Cont.

DL Models	Application in IoMT	Traditional Technology Performance	Enhanced Performance with DL
Gated Recurrent Unit (GRU) [100]	Sequential data analysis in real-time systems	With RNN, the accuracy rate is 88.25% [207]. Traditional method has difficulty in detecting complex patterns	With GRU, 95.15% accuracy rate in predicting heart diseases [207]
Bidirectional Recurrent Neural Networks (BRNNs) [101]	Sequence learning for tasks requiring context from both past and future	Tradition method has difficulty detecting complex patterns	95.47% accuracy rate [208]
Graph Neural Networks (GNNs) [102]	Modeling relationships between entities in medical data	Manual analysis, long R&D cycles	Accelerated drug discovery with AI-driven insights, 99.98% accuracy [209]
Generative Adversarial Networks (GANs) [103]	Data augmentation, synthetic image generation	High radiation dose required for clarity	4x resolution improvement with reduced radiation [210]

5. Challenges

We now summarize some challenges based on our study.

5.1. Security Issues of DL Applications in IoMT

Data privacy and security: Smart healthcare faces significant data privacy and security challenges due to sensitive personal health information from IoT devices and wearables. These sources are prime targets for cyber attacks. With more connected devices, healthcare networks are increasingly vulnerable to hacking and data breaches. Many IoT devices lack proper security, risking unauthorized access. Healthcare providers must navigate strict regulatory environments like HIPAA and GDPR [211] to ensure compliance while maintaining robust security. Breaches can lead to severe penalties and loss of patient trust, highlighting the need for innovative security measures in smart healthcare systems to protect sensitive data. Federated learning, combined with DL, offers a solution to solve these challenges. Raw patient data remain on IoMT devices locally, and only the necessary data are shared with a central server. This decentralized paradigm reduces the exposure of sensitive health information to potential attackers.

Attack on service: This attack aims to disrupt the accessibility of systems managing medical data or connected IoMT devices. IoMT devices often have limited specifications, including constrained memory, network bandwidth, and battery life, as well as data storage capacity, making them particularly susceptible to denial-of-service (DoS) threats. Hackers frequently target hospital networks with DoS attacks to affect patient care. These attacks can delay or block access to critical treatments, including life-saving medications, and hinder clinicians from retrieving essential patient records, significantly impacting healthcare delivery [212]. DL can be utilized as an intrusion detection system (IDS) to identify and analyze network attacks, supporting the timely detection of malicious activities and mitigating potential threats to the system.

Hole attacks: In this type of attack, malicious nodes, known as sinkhole attackers (SHAs), lure legitimate nodes by presenting themselves as the most efficient route to a target destination. Once the legitimate nodes start routing their packets through the malicious nodes, the attackers disrupt network traffic in several ways: they may selectively drop packets to avoid detection by an IDS, fail to send critical information to the destination, or transmit incomplete or altered data. The consequences of such attacks include a significant decline in network performance, reduced communication reliability, and overall inefficiency in data transmission [213]. DL models, such as CNN and RNNs, could be trained to recognize patterns of normal network behavior. Any deviation from expected routing

behaviors, such as excessive packet drops and unusual route changes, can be flagged as a potential hole attack [214].

Attack by a blackhole: Within this attack, a malicious node, known as a blackhole, discards all packets received from its neighboring nodes that are intended for further transmission. The attack becomes even more severe when the blackhole node acts as a sinkhole, disrupting all data flow in its vicinity. Consequently, communication in the affected area halts entirely. This type of attack is often referred to as “selfishness” in the literature [215]. Since blackhole attacks manipulate network routing, GNNs could be leveraged to analyze network topologies and detect inconsistencies in routing behavior. When node traffic is missing, it will be flagged as suspicious [214,216].

Grayhole attack: This variation of the blackhole attack, often referred to as “selective forwarding” or “select and forwarding”, involves a rogue node selectively dropping only a portion of the packets it receives. This strategy makes identifying the attack challenging for an IDS. In multi-hop networks, packet forwarding is a crucial responsibility of routing nodes. However, malicious nodes intentionally discard specific messages in a selective forwarding attack, ensuring they are not passed to neighboring nodes. Unlike traditional blackhole attacks, which risk detection as neighboring nodes may seek alternative routes, the grayhole attack mitigates suspicion by forwarding some of the traffic to neighboring nodes, thereby reducing the likelihood of being exposed [217]. A wormhole attack occurs when an attacker creates malicious connections in the network, enabling it to transmit packets faster than the usual data transfer rate. This creates a “wormhole” in the network, leading to significant disruption [136]. Since grayhole attacks involve selective packet drops, a traditional rule-based IDS may struggle to detect them. DL-based IDS models can learn subtle traffic anomalies and flag suspicious nodes. For instance, DL can be used to train network nodes to bypass grayhole attackers by learning the safest routes and rewarding nodes that maintain a high packet delivery ratio while penalizing malicious nodes [218].

Replay attack: Within this attack, the attacker uses redirection to steal or intercept transmitted information, which can have severe consequences for systems like medical networks. The attacker records the intercepted packets and then “plays back” to the recipient at a later time. This attack can result in two primary outcomes: theft of sensitive information and unauthorized access to specific medical systems. The core strategy involves deceiving the recipient by storing data stored by a malicious node without authorization and retransmitting them as if they originated from the legitimate sender. The malicious sensor can record network traffic to impersonate the original sender and connect with the recipient. This attack is particularly effective in bypassing authentication mechanisms, such as certificates. Even in cases where communications are encrypted, the attacker can gain network access by resending legitimate connection messages without needing to know the actual keys or passwords [33]. Similar to the mechanisms employed for mitigating hole attacks, DL can enhance security against replay attacks by leveraging rapid anomaly detection, pattern recognition, and adaptive security strategies to identify and prevent unauthorized retransmission of data packets.

Sybil attack: A Sybil attack occurs when a node unlawfully assumes multiple identities, disrupting the network’s routing protocol. These attacks compromise data security, resource utilization, and system integrity. Using either a newly created or a hijacked legitimate identity, the Sybil node can communicate with nearby nodes, leading to a disorganized network that may eventually collapse [219]. Additionally, in typical network scenarios, sensors can fail due to power depletion. Exploiting this, an attacker can replace the sensor with a malicious one to perform harmful activities. This can include modifying patient data or introducing false information into the system [220]. DL can significantly enhance

security by recognizing and preventing unauthorized access through behavior analysis, pattern recognition, and adaptive security mechanisms.

Table 11 summarizes various types of attacks on DL-based IoMT networks; their descriptions and corresponding solution methods are given.

Table 11. IoMT network attacks and solution methods.

Type of Attack	Description	Solution Method
Data poisoning	Injecting malicious data into the dataset used for training to degrade model performance.	Implement robust data validation and use differential privacy during training.
Eavesdropping	Interception of data during transmission between IoMT devices and servers.	Use encryption for secure end-to-end data transmission.
Adversarial attacks	Manipulating inputs to cause DL models to make incorrect predictions.	Deploy adversarial training and use input sanitization techniques.
Denial of service (DoS)	Flooding the network with traffic to disrupt services and disable IoMT devices.	Implement network traffic monitoring and distributed denial-of-service (DDoS) mitigation strategies.
Man in the middle (MitM)	Intercepting and altering communication between IoMT devices.	Use mutual authentication and encryption for secure communication.
Model inversion	Reconstructing sensitive training data from the output of learning model.	Apply differential privacy and limit access to model outputs.
Membership inference	Determining if specific data points were used during training.	Use privacy-preserving techniques like federated learning and noise injection.
Ransomware	Encrypting data or systems and demanding payment for decryption.	Maintain regular backups and implement strong access controls.

5.2. Interoperability

Healthcare ecosystems struggle with interoperability between devices, systems, and software platforms. Legacy systems, incompatible with newer technologies, make integration difficult. Proprietary communication protocols complicate efforts to enable devices and platforms to work together. This results in data fragmentation [221], which reduces the effectiveness of patient care. Inconsistent adoption of standardization efforts such as HL7 and FHIR [222] leaves gaps in system compatibility, which pose a significant barrier to adopting smart healthcare technologies.

There are four basic levels of interoperability [47,223,224], as indicated in Table 12.

However, existing healthcare systems were not designed for Level 3 IoMT devices. Integrating DL-powered IoMT solutions into legacy infrastructure requires substantial investment, staff retraining, and overcoming technical barriers [223]. In the IoMT, there are concerns about bias in DL models, especially when training data are not diverse. Ethical dilemmas arise when using AI to make critical healthcare decisions. Legal frameworks lag behind technological advancements, leading to ambiguity in accountability and liability. Integrating DL-powered IoMT solutions into existing healthcare infrastructures requires substantial financial investment to upgrade legacy systems, retrain staff, and address technical limitations, and to ensure seamless data exchange between devices, hospital information systems, and AI models. Key standards concentrate on three areas that are widely adopted in the IoMT area. These include (i) Fast Healthcare Interoperability Resources (FHIRs) [225]: A modern, API-based standard for enabling the sharing of electronic health records (EHRs) between systems. (ii) Health Level 7 (HL7) [226]: A widely used framework for exchanging, integrating, and retrieving electronic health information. (iii) Digital Imaging and Communications in Medicine (DICOM) [227]: The standard for medical imaging data interoperability, including X-rays, MRIs, and CT scans. (iv) Integrating the Healthcare Enter-

prise (IHE) [228]: A framework that integrates existing standards like HL7 and DICOM for seamless healthcare IT communication. (v) ANSI ASC X12 [229]: A standard for electronic data interchange (EDI), commonly used in insurance claims and billing.

Table 12. Basic levels of interoperability.

Level Name	Description	IoMT Application Examples
Level 0 (no interoperability)	Independent systems that lack interoperability.	Stand-alone systems.
Level 1 (also called technical interoperability)	It employs communication protocols to facilitate data exchange between two systems, emphasizing the physical and network infrastructure for data transmission between devices or systems [224].	A wearable heart monitor sending data to a smartphone app via a secure Bluetooth connection.
Level 2 (syntactic interoperability)	It enables two or more systems to transmit data and share services through a common interoperability protocol (e.g., the High-Level Architecture (HLA) [224]).	A glucose monitoring device and an insulin pump to exchange data using a standardized messaging format like JSON, XML, or HL7 (Health Level Seven).
Level 3 (semantic interoperability)	Two systems or more interpret information to generate results that are specified by end-users [223].	A heart rate monitor sends a patient's data to the hospital's EHR system. Semantic interoperability ensures that the EHR understands data meaning as well, such as the EHR can trigger an alert to the healthcare providers if the heart rate is over 120 bpm.

Security and privacy standards related to IoMT include the following. (i) Health Insurance Portability and Accountability Act (HIPPA) [230]: A regulation in the USA to ensure patient data privacy and security. (ii) General Data Protection Regulation (GDPR) [231] European regulation for protecting patient data in healthcare applications. (iii) ISO27001 [232] A standard for managing information security, applicable to healthcare IT. (iv) ISO 27799 [233]: A global set of guidelines for information security in healthcare organizations. (v) OAuth 2.0 [234] & OpenID Connect [235]: Aiming to enable secured authentication and authorization in healthcare applications.

The most important medical device and IoMT standard is IEEE 11073 [236], a standard for network communication between medical devices and healthcare IT systems. According to device deployment with DL, we match the above standards with DL applications to demonstrate how DL can be utilized in the interoperability area. Using wearable devices as an example, DL models such as CNN, RNN, and LSTM are used to detect abnormal activities or predict the potential risks, as Table 13 indicates.

Beyond technical challenges, there are significant ethical and legal concerns. Bias in DL models, often stemming from non-diverse or unrepresentative training datasets, raises questions about fairness and equity in healthcare delivery, particularly when AI is involved in critical decision making. This bias can lead to misdiagnoses or unequal treatment outcomes for specific populations, undermining trust in AI-driven systems. Moreover, ethical dilemmas arise when AI overrides human judgment or operates without clear accountability frameworks. Legal frameworks have not kept pace with rapid advancements in AI and IoMT, resulting in ambiguities around liability and accountability in cases of system failures or adverse outcomes. These challenges underscore the need for cohesive standardization, improved data interoperability, and comprehensive governance frameworks to enable integrating AI and IoMT into modern healthcare systems.

Table 13. Deep learning applications in IoMT with standards.

Category	Standards	DL Applications (Use Cases)	DL Models
Wearable Devices	IEEE 11073 [236], FHIR [225], HL7 [226]	Wearable devices, such as smartwatches and biosensors, detect ECG abnormalities, etc. [25,126].	CNN, RNN, LSTM
	IEEE11073, FHIR, ISO/IEC 27001 [232]	Wearable devices for fall detection [171].	CNN, YOLO
	DICOM [227], FHIR, HL7	Sleep apnea detection analyzes breathing patterns in sleep monitors [158].	LSTM, Autoencoders
Hospital Devices	ISO 27799 [233], NIST Cybersecurity Framework, HIPAA [230], GDPR [231]	Detection of cyber threats such as Sybil, replay, blackhole attacks, etc.	GAN, CNN, Autoencoders
	ISO/IEC 27001, OAuth 2.0 [234], HIPPA	Biometric authentication for EHR access.	CNN, DeepFace
	FHIR, ISO/IEC 27001, GDPR	Federated learning for secure IoMT, keep data locally.	Deep learning embedded in IDS
Home Devices	IEEE11073, FHIR, HL7	Elderly care, smart sensors monitor daily activity, detect abnormal activities, and send alerts [146,147].	CNN, YOLO, Transformer-based models
	FHIR, HL7, ISO27799	Voice assistants, such as detecting speech changes and assisting with medication reminders [54].	RNN, LSTM, Wav2Vec

5.3. Scalability

As healthcare systems incorporate an increasing number of devices and data sources, ensuring scalability and maintaining system performance has become a critical challenge [237].

Currently, sensors and wearable devices connect real-time data and then transmit the data through a central device, which is called a gateway. The gateway filters the data and transmits only the required data to the central cloud application. Users like patients, doctors, and healthcare providers can access, retrieve, or even monitor the changes through cloud applications to obtain the information they are looking for at any time and from anywhere [238]. However, the rapid proliferation of interconnected IoT devices, networks, and sensors introduces a level of complexity that makes scaling infrastructure without performance degradation particularly difficult. The significant influx of data generated by thousands of IoT devices often overwhelms existing systems, creating bottlenecks that can hinder real-time processing and decision making. Furthermore, resource limitations in cloud and edge computing infrastructures exacerbate these challenges, leading to slower response times, increased latency, and overall inefficiencies in system operations. These inefficiencies not only compromise the quality of patient care but also impede effective data management and timely decision making. Abbas et al. designed a framework called BSDMF as an example [239]. According to their experimental results, their framework, including blockchain with distributions, realized a high accuracy ratio and rapid response time. However, their experiment only consists of 100 nodes, from which the conclusion cannot be drawn that the scheme is scalable enough to satisfy expectations.

Another challenge related to scalability is using the IoMT with a blockchain. A blockchain is used to protect communication between devices with smart contracts between nodes and proof of work (PoW) to validate the transaction. It is a decentralized distribution technology that enables a peer-to-peer network to increase not only privacy but also scalability [240]. However, applying a blockchain might increase the response time because of the protections applied on top of the data message before transmission. In addition, the

access-control list (ACL), which is normally applied in the central server to control the devices, might not be applicable with the introduction of a blockchain, which applies to the P2P instead of server mode [240].

To address these challenges, scalable smart healthcare systems must adopt robust architectures capable of handling growing demands while ensuring performance, reliability, and security. Cloud-based models are essential for centralized processing of large datasets, enabling global access and resource allocation. Similarly, edge computing offers localized data processing closer to the source, reducing latency and optimizing resource usage [241]. For instance, Mallick et al. proposed a framework called EMRI to add the smart contract interface before the smart contract to collect and filter the data from various IoMT devices, then use Blockchain's smart contract to send information to the central server [240]. With their proposal, the blockchain technology is only used between the interface to the central server rather than between nodes. With their proposal, the communication between nodes is eliminated since communication between nodes is not necessary. However, their proposal adds single-failure risks. If there is anything wrong in the interface area, all the information might be lost before going to the smart contract.

The integration of these models, alongside advanced optimization techniques, can help healthcare systems manage the exponential growth in data and device connectivity, ensuring both scalability and high-quality care.

5.4. Data Quality and Management

The healthcare sector faces significant challenges in managing data quality [242] due to the inherent complexity and diversity of data from wearables, electronic health records (EHRs) and patient monitoring systems. Poor data quality can result in serious consequences, including misdiagnoses, erroneous analyses, and inappropriate treatment plans. Data collected from IoT devices and wearable sensors often contain noise, which obstructs meaningful insights. Furthermore, inconsistencies in data collection processes complicate analysis, while missing data introduce critical gaps in patient records, ultimately compromising the quality of care. The overwhelming volume of data also contributes to clinician fatigue, which can lead to suboptimal decision making. Adding to these challenges is the lack of standardized datasets, evaluation protocols, and interoperability among IoMT devices, which creates a significant barrier to scaling IoMT solutions. Variations in data formats, sensor specifications, and communication protocols result in ecosystem fragmentation, making it difficult to achieve seamless integration. Addressing these issues requires the development of globally recognized standards for datasets and evaluation metrics.

Standardization efforts will not only improve interoperability but also facilitate the fair benchmarking of DL models, driving more reliable and scalable IoMT applications. For instance, Apple Inc. proposed platforms called ResearchKit and CareKit as data brokers to standardize the format of data that are collected and processed for healthcare records through Apple's devices [243].

AI has been proposed to standardize the data format as well. For instance, in healthcare, 10MT data aggregation refers to collecting and analyzing data from 10MT devices to review a patient's health [244]. Such 10MT data from 10MT devices include data such as biometric information, clinical information, and data from medical devices themselves [244]. ML plays a key role in the 10MT data aggregation, such as predicting the results based on the data collected. However, ML faces challenges in obtaining various data formats collected from multiple sources. An ML score system can be added to evaluate the data collected before processing the data. In this, any data with a low score will be highlighted and evaluated before moving to the next step [244].

Historical data are always used to train ML models. However, lots of historical data might include incomplete information without a standard data structure. In order to solve this issue, ML or AI models can be added to deal with these incomplete data [245]. Also, AI tools to convert data into a consist format not only improve overall system performance but also simplify integration within healthcare systems like electrical health records (EHRs). This also ensures regulatory compliance and enables local response for prompt decision making.

5.5. Real-Time Processing and Energy Efficiency

Many IoMT applications, such as detecting abnormalities in ECG or oxygen levels, require instantaneous responses. Ensuring ultra-low latency in processing and delivering predictions is technically challenging, especially in cloud-dependent setups. Devices, especially wearables and implants, operate on limited battery life. Continuous running of DL algorithms can drain energy quickly, limiting device functionality. Balancing model complexity and energy efficiency is a critical challenge. Real-time processing and energy efficiency are critical challenges for IoMT applications, such as detecting abnormalities in ECG patterns, oxygen levels, or vital signs. These use cases demand ultra-low latency to ensure timely interventions, but reliance on cloud-based architectures often introduces delays due to data transmission and processing. Additionally, wearable and implantable IoMT devices operate on limited battery life and computational resources, making the continuous execution of DL algorithms highly energy-intensive.

Complex DL models, such as those with convolutional layers or transformers, provide high accuracy but are impractical for resource-constrained devices without optimization. Strategies to deal with these challenges include leveraging edge computing to perform local data processing and reducing latency and energy consumption. Model optimization techniques, such as 10MT aggregation to pruning, quantization, and knowledge distillation, minimize computational demands while maintaining accuracy [244].

Federated learning enhances privacy and reduces communication overhead by enabling collaborative training without transmitting raw data to the cloud. For instance, Ni et al. designed a federated split learning (FedSL) framework to collaboratively use healthcare analytics on a number of IoMT devices with limited resources [246]. Arya et al. proposed to use FL to train a deep learning (DL) model to classify decentralized data from various IoMT devices [247]. Likewise, Gupta et al. proposed an IoMT framework named “FedCare” to add edge nodes to standardize the data to solve heterogeneity issues in federated learning [248].

Energy-efficient hardware, including specialized GPUs and TinyML platforms, along with adaptive algorithms that adjust model complexity based on resource availability, further improve efficiency. Demirel et al. presented an energy-efficient design approach for monitoring the heart from wearable devices continuously. According to their proposal, a hybrid edge–fog/cloud IoMT system is proposed to detect noise before transforming the data to save on the overall energy assumption [249]. Soman et al. proposed to use the Coral Reef-Based Energy-Efficient Framework to save energy. In their research, a coral means a number of coded solutions. A special grid cell is used to store the data. If better data come out but no space is available in the reef, the new data will replace the old data. Such a method reduces the overall size to reduce the assumptions during the data processing [250].

Incorporating low-power communication protocols like Bluetooth Low Energy (BLE), and advancements in battery technology, such as solar-powered or rechargeable batteries, can extend device longevity. For example, Arslan et al. used BLE 5.0 on a wearable wristband to send data to the server [251]. However, BLE is subject to man-in-the-middle (MITM) attacks during the pairing called “Just Works” which does not provide any I/O capabilities [252,253]. In order to protect devices during the pairing process, security methods such as ACL and blockchain, should be considered to add on top of the BLE 5.0. Salem et al. proposed the Ephemeral Elliptic Curve Diffie–Hellman (EECDH) to obtain a session key to provide protection on the BLE5.0 [251]. Locality-sensitive hashing (LSH) is used to identify nearest neighbors in ML and DM to ensure only nearby data are reviewed. LSH is widely adopted to identify abnormal measurements or those deviating from the hash values; this can be applied on top of BLE to ensure only authorized devices are accepted to retrieve the data [251].

These approaches enable IoMT systems to satisfy the dual demands of timely data processing and energy efficiency, ensuring reliable, scalable, and effective smart health-care solutions.

Table 14 compares DL models’ performance, concentrating on the accuracy, computational overhead, model size to treat, the spread of processing, and generalization ability. When calculating the overhead performance, we adopted floating-point operations per second (FLOPs) to represent the number of floating-point arithmetic operations a model needs to perform during training. FLOPs are commonly used to assess the efficiency of deep learning models. Based on our findings, GANs have the highest overhead, requiring high-end GPUs for clinical text processing, which explains why GANs are widely used in cloud computing rather than local devices. DNNs and RNNs have moderate to high FLOPs, making them ideal for predictive modeling and real-time monitoring. Their speed is relatively slow compared to the model size they are processing. Therefore, they are effective for local applications with small model sizes. Another finding is that the latest DL model, MobileNetV3, is highly efficient for real-time health monitoring, with an inference time of only 10 ms. According to a study by [254,255], using MobileNetV3 not only shortens the time but also improves the accuracy rate by around 1.2% compared to other CNN models such as Visual Geometry Group (VGG). However, MobileNetV3’s accuracy rate suddenly drops when dealing with complex image scenarios, and it is sensitive to small datasets, and so may not be optimal for extremely high-resolution image processing due to its design focus on lightweight operations. Therefore, MobileNetV3 is less suitable for tasks requiring detailed feature extraction in large images [255]. To ensure a high accuracy rate with stable performance, VGG-16 is preferred over MobileNetV3 [256]. For mobile personal devices, such as smartwatches, with limited memory, MobileNetV3 is preferred over VGG-16 due to its smaller model size and efficiency.

Table 14. DL processing and energy efficiency performance.

DL Model	Application in IoMT	Dataset	Accuracy (%)	Overhead (FLOPS)	Model Size (MB)	Inference Time (ms)	Generalization Ability
Convolutional Neural Networks (CNNs) [93–95]	Image analysis (e.g., classification, segmentation, detection)	Chest X-ray	Accuracy: COVID-19 CT images detection: 97.0% with VGG19 algorithm [202]	High (4.1 G) [95]	25–90 [95]	50–100 [162]	High [257]
Deep Neural Network (DNN) Transformer Models [18]	Multimodal input fusion (e.g., imaging data, clinical data)	Disease risk prediction	Around 89% [94]	Medium (2.5 G)	10–50	30–100	Moderate [257]
Recurrent Neural Networks (RNNs) [96,97]	Time-series analysis (e.g., vital signs, ECG, EEG data)	MIT-BIH	90%+ accuracy in anomaly detection, real-time alerts [205]	Medium (2.8 G)	20–50	30–100	Moderate [257]
Long-Short Term Memory (LSTM) [98,99]	Time-series data analysis (e.g., ECG, EEG, blood pressure monitoring)	ECG	Accuracy of 97.6% [206]	Medium (2.8 G)	20–50	30–100	Moderate [257]
Gated Recurrent Unit (GRU) [100]	Sequential data analysis in real-time systems	MobiAct	95.15% accuracy rate to predict heart diseases [207]	Medium (2.8 G)	20–50	30–100	Moderate [257]
Bidirectional Recurrent Neural Networks (BRNNs) [101]	Sequence learning for tasks requiring context from both past and future	PhysioNet	95.47% accuracy rate [208]	High (5–6 G)	20–80	50–120	High [257]
Graph Neural Networks (GNNs) [102]	Modeling relationships between entities in medical data, such as drug discovery and personalized medicine.	DrugBank	Accelerated drug discovery with AI-driven insights, 99.98% accuracy [209] Average is around 87.5%	High (5.4 G)	30–120	100–250	Low (task dependent)
Generative Adversarial Networks (GANs) [103]	Data augmentation, synthetic image generation	NIH Chest Ray	88.3%	High (6.5 G)	100–300	150–300	Low (domain-specific)
MobileNetV3 [258]	Exact images	COVID-CT and COVID-XRAY	1.2% higher than other CNN models [254]	Low (0.3 G)	14	10	High [254]

6. Opportunities

Addressing challenges and pursuing future research directions will enable the field of DL in IoMT to continue evolving, providing more effective, reliable, and secure solutions for smart healthcare.

Performance scalability and computational complexity: The implementation of DL in IoMT systems is constrained by significant challenges related to scalability and computational complexity. As IoMT applications become increasingly sophisticated, such as those involving robotic surgery, remote patient monitoring, or multimodal diagnostic systems, integrating diverse data sources and the hierarchical nature of advanced DL models create immense computational demands. Robotic surgery, for example, requires real-time processing of high-resolution imaging data, precise control of robotic instruments, and seamless integration of multiple sensors and feedback loops. These operations often require powerful hardware and substantial computational resources, which may not be readily available in resource-constrained clinical settings. In addition, scaling these models to accommodate increasing data volumes and connectivity between hospital networks or home healthcare systems presents additional challenges. Current DL architectures, while accurate, are often computationally expensive and not optimized for applications with real-time requirements.

Future research must address these issues by simplifying model architectures through various optimization techniques. For instance, using model quantization and pruning to reduce memory and computation costs, as seen in MobileNet [259] for edge devices. Knowledge distillation, as in TinyBERT [260], transfers knowledge from large models to smaller ones while maintaining accuracy. Specialized hardware like Google's TPUs [261] can be leveraged to speed up tasks. Scalability can be enhanced with model parallelism, such as distributing tasks to multiple GPUs for efficient training. Federated learning can be used to enable local device training without sending sensitive data to central servers. Asynchronous training methods, such as those used in Horovod [262], reduce synchronization bottlenecks in cloud environments. To reduce complexity, SparseGPT removes redundant parameters to enhance efficiency [263]. These techniques make DL models more practical for real-world applications, from edge AI in IoT devices to large-scale cloud training for smart agriculture.

Data quality improvement: A critical challenge in applying DL in smart healthcare CPSs is the lack of standardized datasets and evaluation protocols. This gap hinders the ability to systematically compare the performance of different models and studies, creating inconsistencies in reported results. For example, in atrial fibrillation detection, variations in data collection methods, labeling standards, and dataset composition often lead to discrepancies in model evaluations. Furthermore, while many DL models demonstrate impressive performance in controlled experimental settings, they frequently struggle to generalize to real-world populations because of the diversity and complexity of clinical data. Factors like demographic variations, comorbidities, and inconsistent healthcare practices exacerbate these issues, limiting the robustness and applicability of these models.

To overcome these challenges, future research must prioritize the development of standardized benchmarks and datasets tailored to the unique requirements of healthcare applications. Establishing uniform data collection and annotation protocols will facilitate fair comparisons and encourage more consistent model evaluations. Furthermore, testing DL models in diverse real-world environments is essential to assess their reliability and generalizability across different populations and healthcare settings. Collaborations among academic institutions, healthcare organizations, and industry stakeholders can further accelerate the creation of comprehensive and diverse datasets. These initiatives will not

only enhance the quality of healthcare data, but will also strength the robustness, reliability, and equity of DL applications in smart healthcare.

Future engineering techniques, such as principal component analysis (PCA) [264], refine datasets by selecting the most relevant attributes, improving model efficiency in financial fraud detection. Active learning, as used by self-driving car systems, prioritizes the most informative data points for labeling, minimizing annotation costs while maintaining model robustness [265]. Missing values can be handled through imputation techniques—such as using K-nearest neighbors (KNN) [266] to ensure complete and reliable datasets. Data consistency and deduplication are crucial in healthcare AI. Bias mitigation strategies, including re-sampling techniques and adversarial debiasing, promote fairness in AI. These techniques collectively improve DL models' accuracy, generalizability, and reliability, making them more effective in real-world applications.

Reliability and resilience: Reliability and resilience are critical considerations for deploying DL models in smart healthcare, particularly in high-stakes and critical care environments. A persistent challenge in DL is overfitting, which occurs when the model performs well on training data but the trained model fails to generalize when dealing with new unseen data. This problem is especially problematic in healthcare care, where limited and non-representative datasets are common. Overfitting can lead to unreliable predictions, misdiagnoses, and reduced confidence in AI-assisted decision making. In addition, variability in patient demographics, medical conditions, and healthcare practices require models that can adapt and generalize effectively across diverse populations and surgical procedures.

To deal with these challenges, further research should prioritize the development of advanced training techniques like data augmentation, transfer learning, and federated learning to enhance model robustness and reduce overfitting. Regularization methods and cross-validation strategies can further improve model generalization. Creating synthetic data through generative adversarial networks (GANs) can also help supplement limited datasets, providing more diverse training samples. Robust testing in various demographic and clinical scenarios is also essential to assess the safety and effectiveness of DL models in real-world settings. Ensuring reliability and resilience in DL applications will not only improve patient outcomes but also foster trust in AI-driven smart healthcare systems.

Advanced AI algorithms and techniques: Emerging research trends in DL highlight the potential of federated learning to enhance data privacy and security in smart healthcare by training models in decentralized data sources [267,268]. In addition, there is a growing need for explainable AI models that can provide transparent and interpretable predictions, particularly in medical diagnostics and patient care.

Future research should explore federated learning and develop methods for explainable AI to enhance the transparency and trustworthiness of DL models in smart healthcare. Federated learning enables IoMT devices to train DL models in a collaborative way without sharing raw data, thus preserving patient privacy and security. Lightweight DL models deployed on IoMT devices facilitate local data processing, which reduces latency, as seen in features such as a smartwatch that detects arrhythmias. Additionally, DL-powered IoMT devices assist elderly and disabled patients with tasks such as fall detection and medication reminders, adapting treatments based on monitored health conditions such as tremors in Parkinson's patients.

Personalized models and data fusion methods: Developing personalized ML models tailored to individual patient characteristics holds immense potential for improving the precision and reliability of health monitoring systems. Every patient has unique physiological patterns affected by factors like age, genetics, lifestyle, and underlying health conditions. Standardized models often fail to account for these variations, leading to suboptimal

performance in detecting anomalies or managing chronic conditions. Personalized models, trained or fine-tuned on individual patient data, can significantly improve detection accuracy, provide more reliable health insights, and facilitate targeted interventions.

Future research should study techniques to personalize DL models like transfer learning, where pre-trained models are fine-tuned on patient-specific data, or adaptive learning, which allows models to evolve continuously as new data become available. Leveraging advanced methods like federated learning can also enable personalization while preserving data privacy, ensuring sensitive health information remains secure. Moreover, integrating data from multiple sources, such as photoplethysmography (PPG), electrocardiography (ECG), and wearable sensors, can enhance model accuracy and robustness. Data fusion techniques enable the combination of complementary signals, providing a substantial understanding of a patient's health. For instance, combining PPG and ECG data can improve heart rate variability analysis, while integrating accelerometer data can provide deeper insights into activity and posture changes. These approaches not only improve health monitoring precision but also pave the way for developing genuinely personalized and holistic healthcare solutions.

Cross-domain applications and adaptive learning: Applying advanced AI techniques like deep reinforcement learning (DRL) [269] beyond healthcare opens up new possibilities for leveraging cross-domain insights in areas like industrial automation and smart cities. Also, the systems need to be operated via future wireless communications that integrate with sensors [270], aiming to providing sensing capability using wireless communication signals for the sake of cost reduction. By exploring how AI-powered solutions perform in diverse domains, researchers can identify strategies and innovations that can be adapted to healthcare, enriching the field with novel approaches. For instance, DRL, widely used in robotics and logistics for optimizing processes, can be adapted for tasks like robotic surgery or workflow optimization in hospitals. Similarly, predictive maintenance models from industrial automation can inform healthcare IoMT systems to proactively address device failures, ensuring seamless operation in critical care settings.

Adaptive learning is another promising area, allowing DL models to adjust to continuously evolving conditions. In healthcare, environments are dynamic, with changes in patient demographics, disease prevalence, and treatment protocols, requiring models that can adapt in real time. Techniques like online learning, where models update their knowledge as new data arrive; meta-learning, which enables models to generalize to new tasks quickly; and transfer learning, to reuse the existing learning model can significantly enhance resilience and applicability [271,272]. Adaptive models are particularly valuable in applications such as personalized health monitoring, where physiological patterns evolve over time, or in smart cities, where environmental and infrastructure conditions change dynamically. Future research should emphasize combining cross-domain applications with adaptive learning to create robust, scalable, and versatile AI systems. These systems can bridge gaps across industries while maintaining the flexibility required to address the specific challenges of dynamic environments like healthcare, enhancing both resilience and impact.

Explainable AI in IoMT: DL models are frequently considered as “black boxes”, making it challenging for healthcare professionals to gain insights into what is behind predictions or decisions. This lack of interpretability undermines trust and could limit the adoption of AI in critical medical applications, where transparency is essential for accountability and informed decision making. The need for explainability becomes even more urgent in the context of IoMT systems, which often deal with life-critical data such as ECG signals, oxygen levels, and patient monitoring metrics.

Explainable AI techniques, including SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations), and attention mechanisms, are examples that aim to deal with this challenge. These methods offer insights into which features or data segments most influence the model's decisions, enabling healthcare professionals to validate AI-driven recommendations. For instance, in IoMT systems analyzing ECG data, explainable AI can identify and highlight specific segments of the signal responsible for detecting anomalies, such as arrhythmias, making the results more interpretable and actionable. In high-stakes applications, such as diagnosing life-threatening conditions or managing chronic diseases, explainable AI ensures that clinicians can confidently rely on AI systems without fearing hidden biases or errors. Furthermore, it enhances collaboration between AI systems and medical professionals, bridging the gap between technological advancement and practical usability.

Future research should focus on integrating explainability into IoMT solutions from the outset, making sure that AI-driven healthcare systems remain transparent and trustworthy, as well as being aligned with the needs of clinicians and patients alike. Trust is a major barrier to AI adoption in healthcare, particularly in high-stakes applications, in which incorrect predictions can have severe consequences. Explainable AI (XAI) directly addresses this concern by improving transparency, interpretability, and accountability in DL-based IoMT solutions. By making AI decisions understandable, XAI fosters greater trust among clinicians, patients, and regulatory bodies. XAI enhances trust in DL applications for IoMT by improving decision transparency, bias detection, regulatory compliance, and clinician–AI collaboration. XAI techniques allow medical professionals to understand why a model made a specific prediction, reducing skepticism and improving confidence in AI-driven diagnostics. For example, in automated diabetic retinopathy detection, XAI can highlight affected retinal regions, helping ophthalmologists validate AI-based assessments. Additionally, XAI helps identify biases in AI models, which may arise from imbalanced training data. This could result in disparities in healthcare outcomes. Through offering insights into model decision making, XAI allows healthcare professionals to recognize and mitigate these biases so that fairness can be ensured.

Regulatory compliance is another crucial factor, as frameworks like GDPR and HIPAA require AI-driven healthcare systems to provide explainable justifications for their decisions. XAI supports auditability, helping regulatory bodies assess and certify AI-powered medical devices for safe deployment. In remote monitoring and telemedicine, IoMT applications generate vast amounts of real-time patient data, where XAI ensures that AI models provide interpretable recommendations, allowing clinicians to make informed, confident decisions in virtual consultations. Furthermore, XAI strengthens clinician–AI collaboration by making AI decision processes transparent, enabling clinicians to combine human expertise with AI insights rather than blindly relying on automated predictions. This hybrid approach improves the adoption and acceptance of AI-driven healthcare systems. In high-stakes applications such as diagnosing life-threatening conditions or managing chronic diseases, explainable AI ensures that clinicians can trust AI outputs without fearing hidden biases or errors. Future research should prioritize integrating XAI into IoMT solutions to make AI-driven healthcare systems more transparent, trustworthy, and aligned with clinical needs.

Sustainable AI in IoMT: IoMT devices, such as wearables and implants, face strict power constraints that significantly challenge their usability and longevity. Running computationally intensive DL models on these devices can rapidly drain their limited battery life, restricting their ability to provide continuous monitoring and insights [273,274]. Developing lightweight DL models has become a critical area of focus in addressing these issues. Techniques such as model pruning, which reduces unnecessary parameters; quantization,

which converts models into lower-precision formats; and knowledge distillation, where smaller models learn from larger ones, are key to significantly lowering computational requirements while maintaining accuracy [275,276]. A framework like TinyML [277] is particularly promising as it deploys compact DL models directly on IoMT devices. These models allow real-time data processing and analysis without relying heavily on cloud infrastructure, minimizing energy consumption and latency.

Integrating renewable energy sources, such as solar-powered sensors, into IoMT devices can support prolonged operations, especially in remote or resource-limited settings. Energy efficiency can also be enhanced during the training phase of DL models. Utilizing low-carbon cloud computing or distributed training methods can reduce the environmental impact associated with the substantial computational demands of training large models. By prioritizing sustainability at both the device and system levels, IoMT can achieve prolonged operation, reduced environmental impact, and improved feasibility for widespread adoption. Future research should emphasize these energy-efficient approaches to ensure that IoMT solutions remain viable, scalable, and environmentally responsible.

7. Final Remarks

The integration of DL with the IoMT presents a groundbreaking opportunity to revolutionize healthcare diagnostics, treatment, and patient monitoring. By leveraging advanced AI-driven analytics and real-time data processing, DL-powered IoMT solutions can significantly enhance early disease detection, personalized treatment plans, and operational efficiency. However, major challenges persist, including computational constraints, data privacy concerns, lack of interoperability, and energy-efficient processing. The heterogeneity of IoMT-generated data, combined with the demand for real-time decision making and security, further complicates widespread adoption, particularly in resource-limited healthcare environments.

To accelerate progress in this domain, future research must prioritize key areas for innovation and practical implementation. First, there is an urgent need for the development of lightweight, energy-efficient DL models optimized for edge computing and IoMT devices. Researchers should focus on model compression techniques, including pruning, quantization, and knowledge distillation, to enable high-performance AI on low-power medical devices. Second, ensuring data privacy and security must remain a top priority, with federated learning, homomorphic encryption, and blockchain-based frameworks emerging as promising solutions to secure patient data without compromising AI performance. Moreover, standardization efforts should be accelerated to establish universal IoMT interoperability standards, allowing seamless integration of DL models across diverse healthcare systems. Collaboration between academia, industry, and regulatory bodies will be critical in creating frameworks that ensure transparency, fairness, and trust in AI-driven healthcare applications.

Additionally, researchers should explore XAI techniques to bridge the gap between DL models and healthcare professionals, improving the interpretability, accountability, trustworthiness, and dependability of AI-assisted decision making. From an implementation perspective, IoT developers and AI practitioners should actively work on deployable solutions tailored to real-world clinical environments, addressing latency, bandwidth constraints, and real-time processing challenges. Encouraging publicly available, diverse, and high-quality IoMT datasets for training and validating DL models will further enhance generalizability and fairness across patient populations. In conclusion, while DL-powered IoMT systems offer unparalleled potential, realizing this vision requires collective efforts from researchers, engineers, and healthcare professionals to tackle computational efficiency, privacy, security, and scalability challenges. By prioritizing lightweight AI models, robust

security frameworks, and scalable interoperability solutions, the research community can drive the next wave of AI-driven smart healthcare innovations, ensuring equitable and high-quality medical care for all.

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